

U21BM505

BIOCONTROL SYSTEMS

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UNIT - IV - PHYSIOLOGICAL CONTROL SYSTEMS

Introduction to physiological control systems - Art of modeling physiological systems - Parametric and non-parametric models - Lumped and distributed parametric models - Positive and negative feedback physiological control systems

C04: Describe the concepts of physiological system modelling (Understand)

PHYSIOLOGICAL CONTROL SYSTEMS

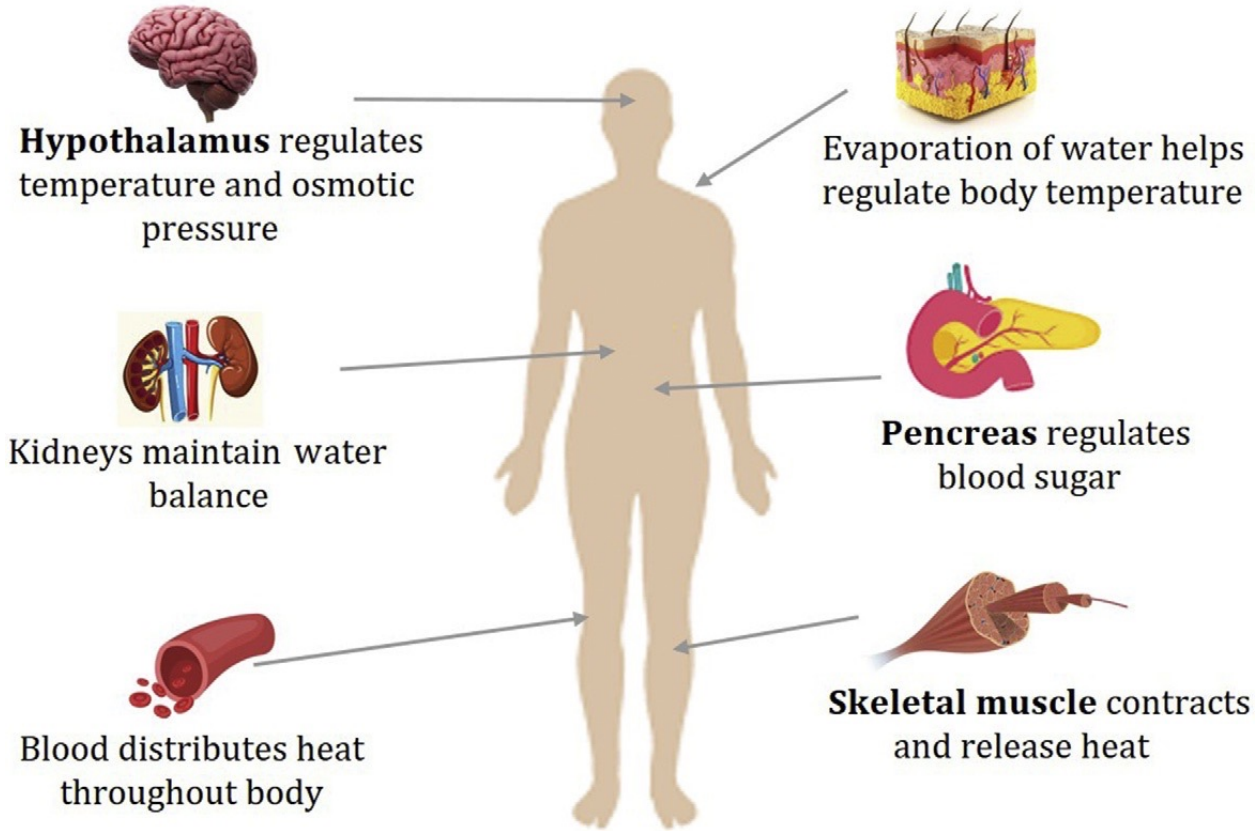
- A physiological model is a mathematical representation that approximates the behavior of an actual physiological system.
- Physiological models can serve mainly for the following purposes:
 - to understand the physiological system
 - to predict its dynamics
 - to control the system under desirable conditions

- In physiology, control refers to the process of stabilizing a physiological variable to a specified set point, either by reversing perturbations via negative feedback closed loops or via anticipatory open loops.
- In the human body, the control process is designed by homeostasis.

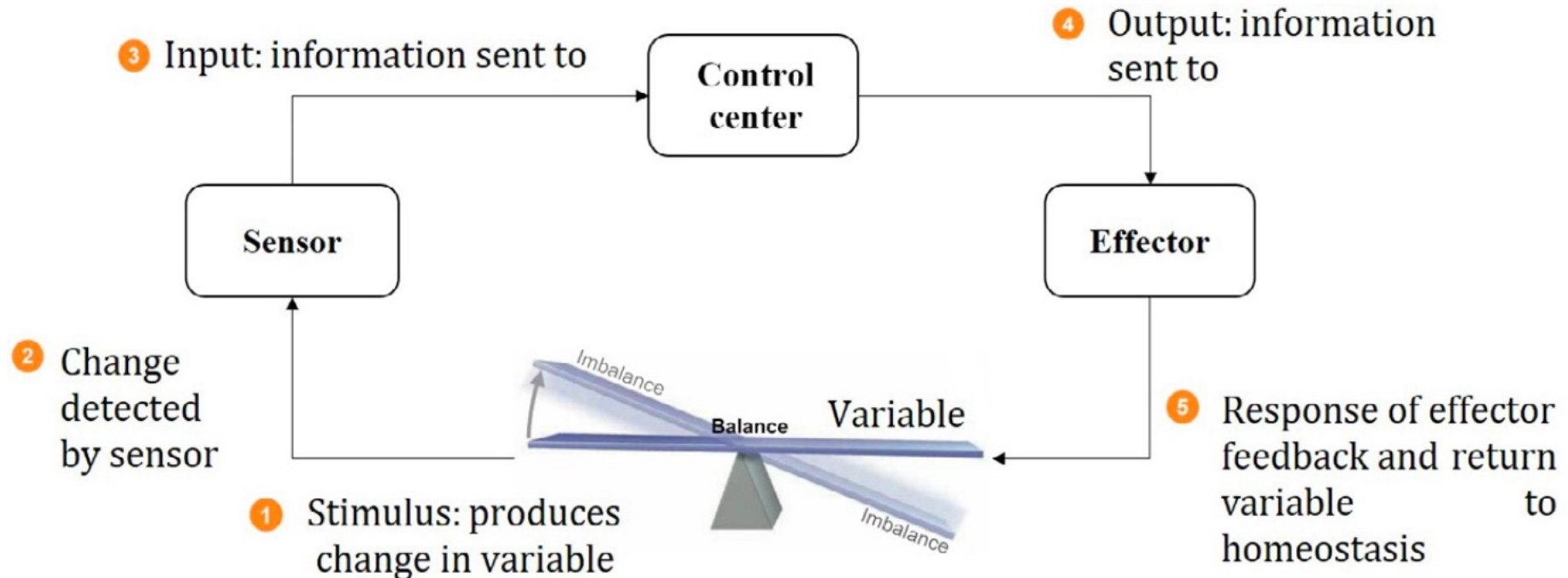
HOMEOSTASIS

- Homeostasis is the process by which an organism maintains a state of balance and stability while adjusting to its environment.
- E.g.: Thermoregulation (sweating cools the body when hot, and shivering generates heat when cold), blood sugar regulation (insulin lowers blood sugar, while glucagon raises it) and pH balance (maintaining a stable blood pH through buffering systems and respiratory adjustments).

Homeostasis

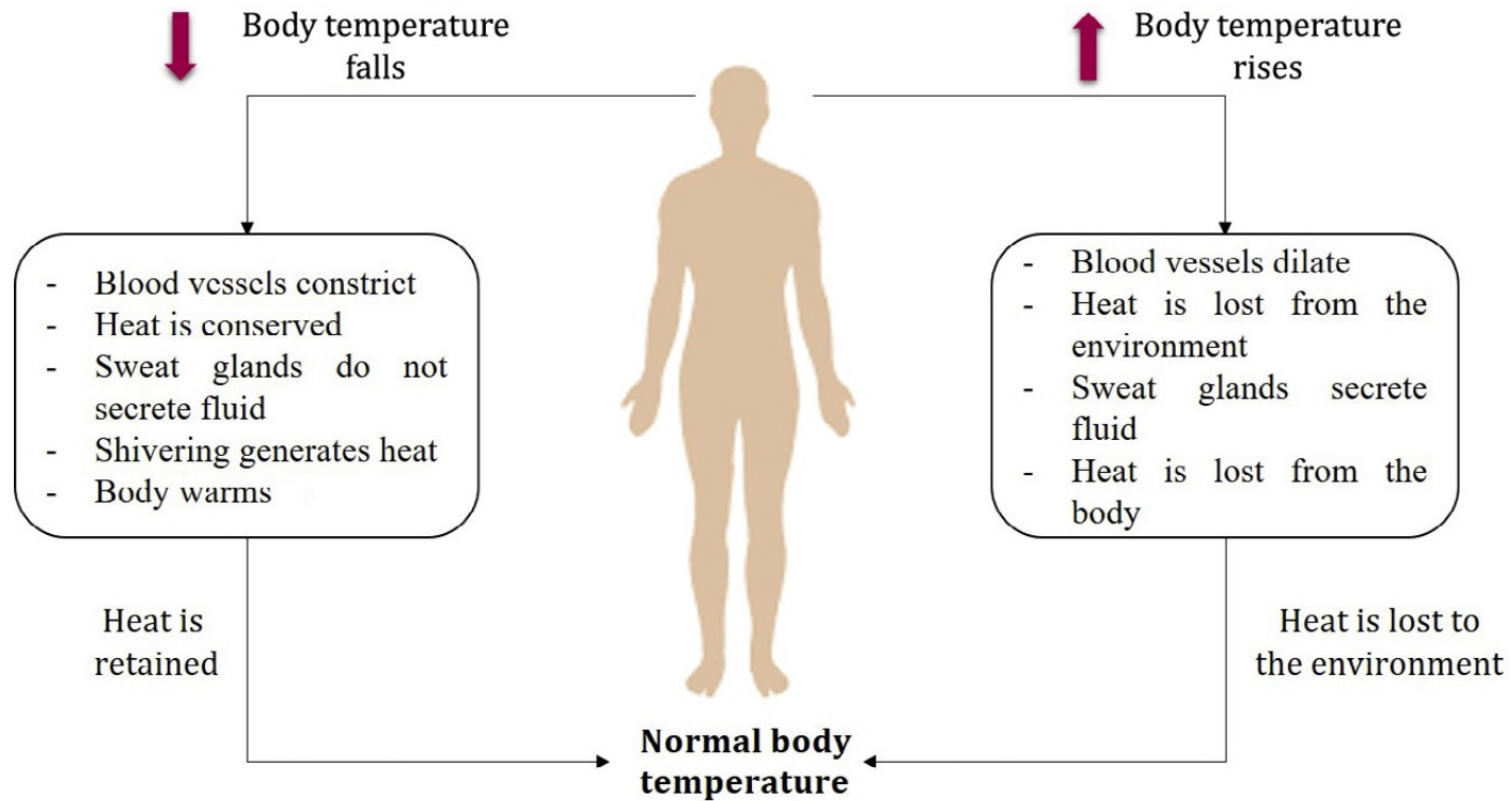


HOMEOSTASIS PRINCIPLE

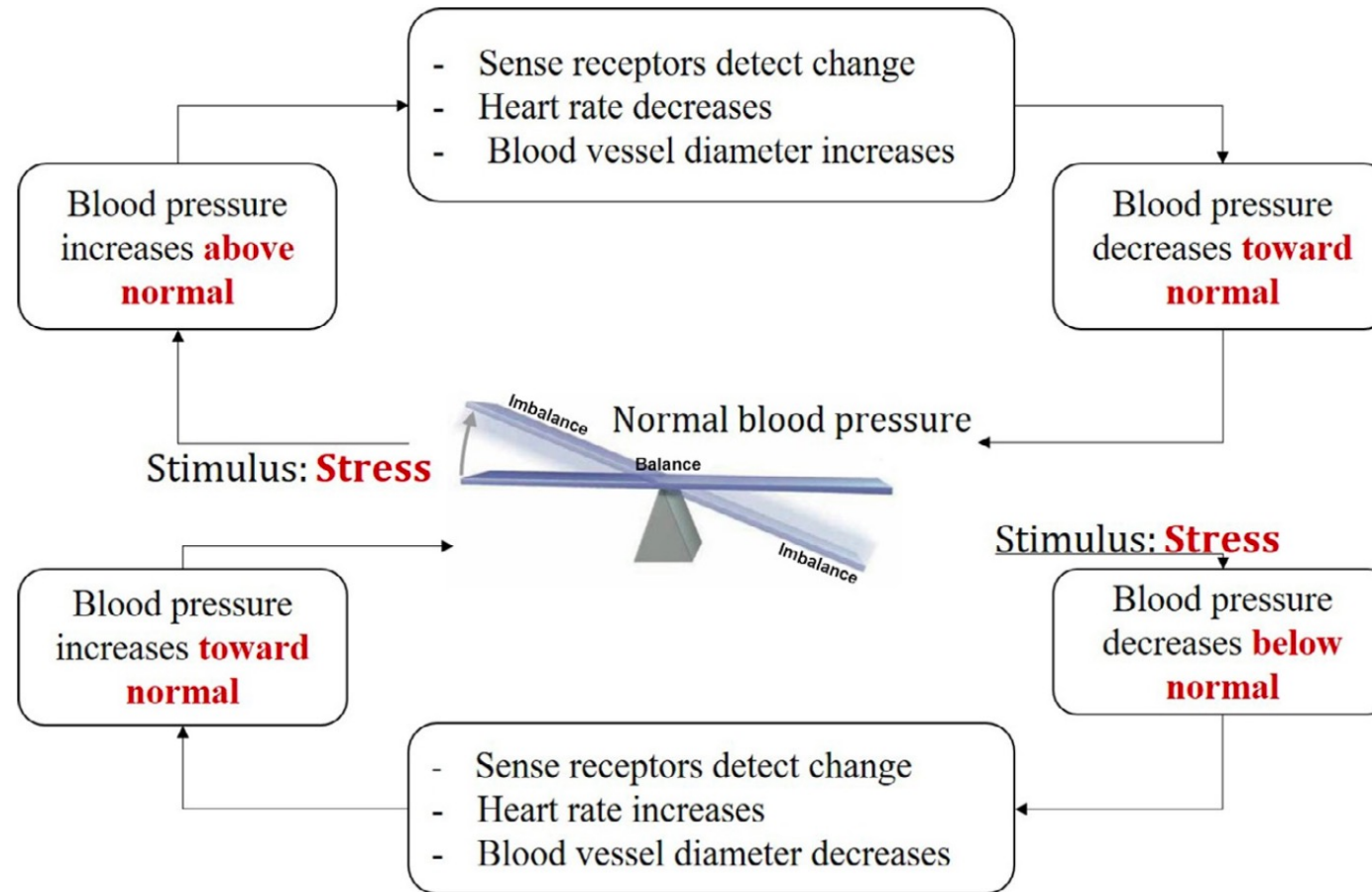


- property of a physiological system to regulate its internal environment to a given set point in presence of a specific stimulus producing changes in that variable.
- objective is to maintain stable and relatively constant physiological behavior.
- control process in the human body is ensured through the coordination of the control center (the nervous system and the endocrine system) and the natural sensors and effectors.

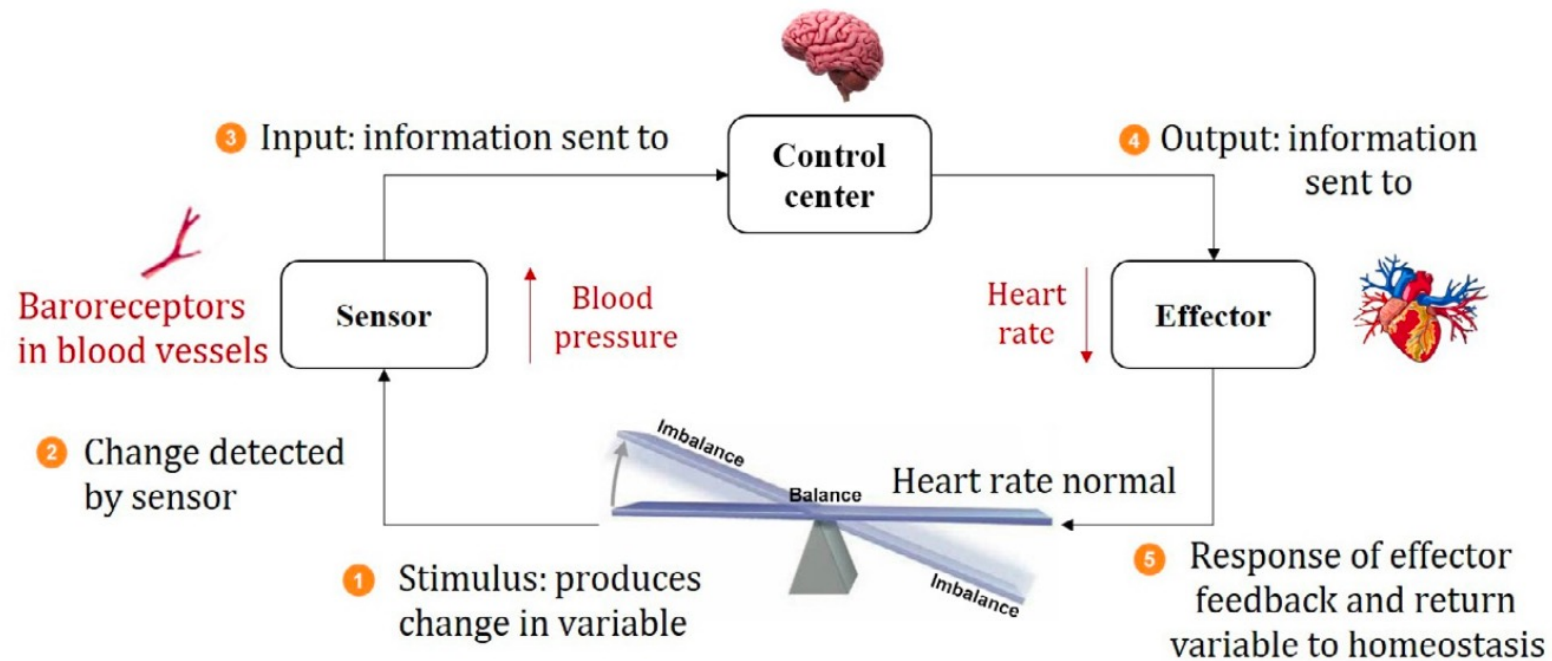
HUMAN TEMPERATURE HOMEOSTASIS



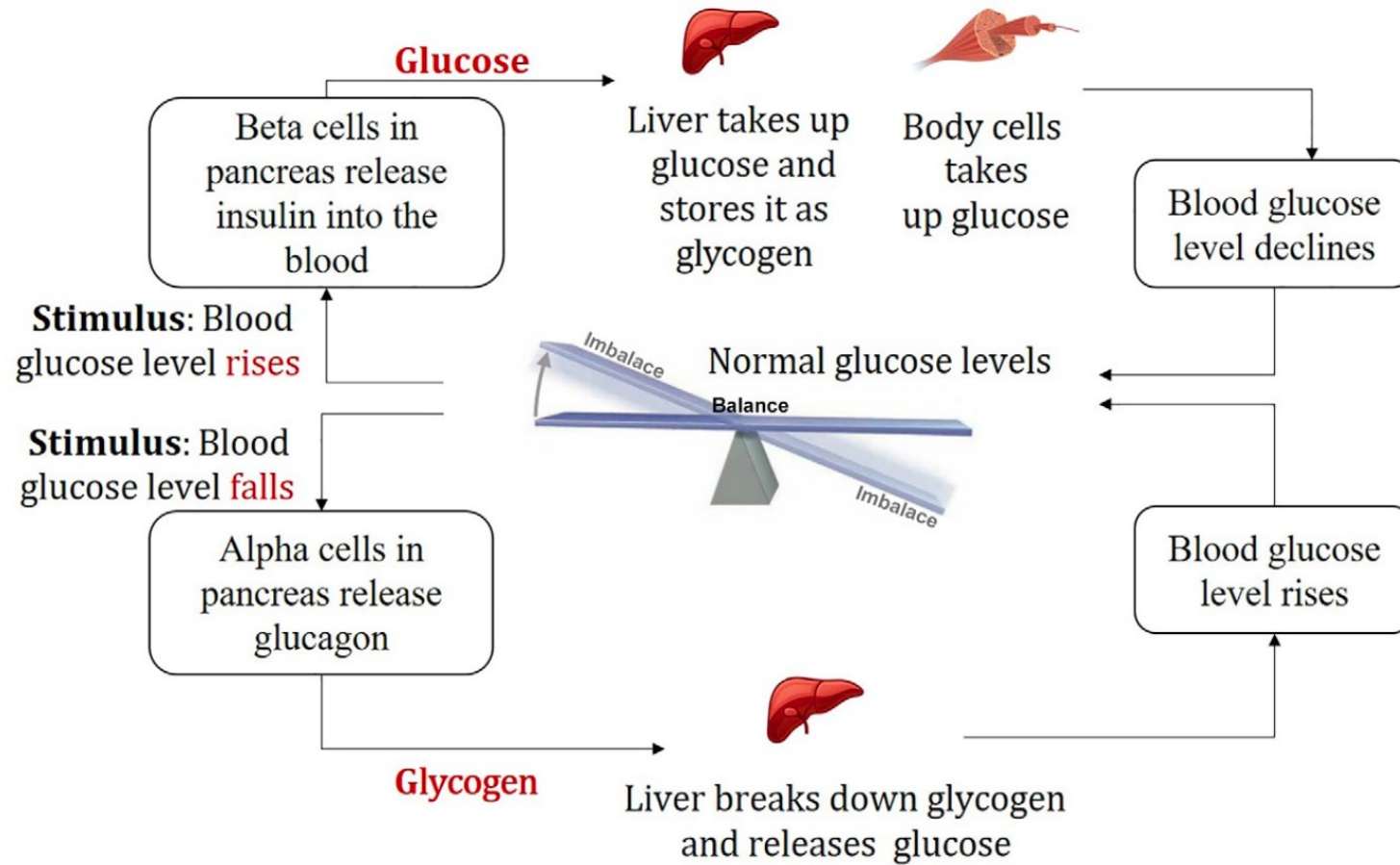
BLOOD PRESSURE HOMEOSTASIS



HEART RATE HOMEOSTASIS

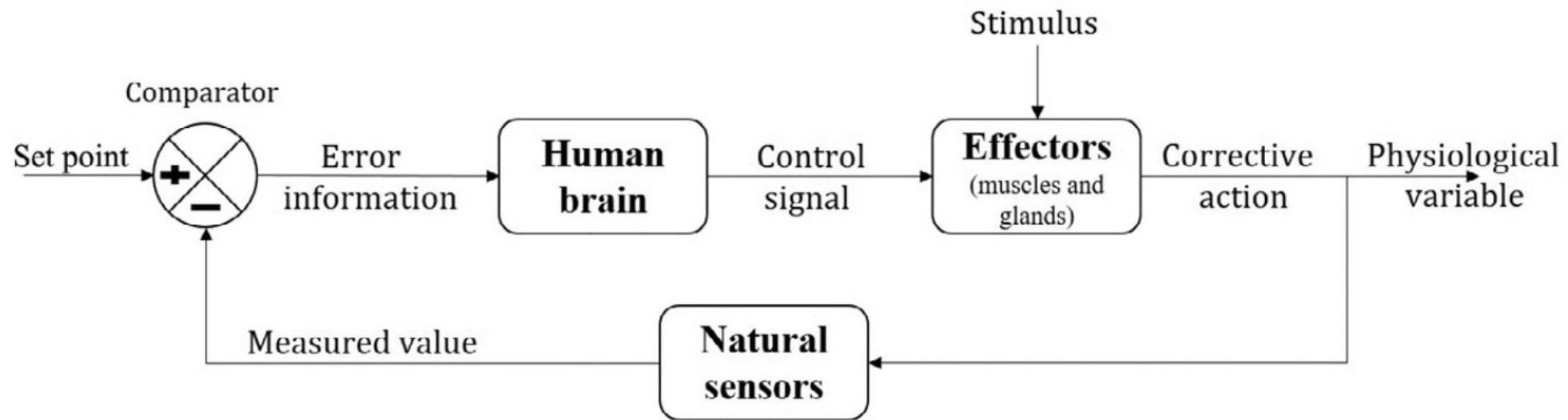


BLOOD GLUCOSE HOMEOSTASIS



COMPONENTS

- Sensor (Receptor): Detects changes in a physiological variable (e.g., temperature, blood glucose, blood pressure). The sensor sends signals regarding the current state of this variable to the control center.
- Control Center (Integrator): Receives input from the sensor, compares it to a set point (the desired value), and decides if action is needed. If there's a deviation from the set point, the control center generates an output signal to correct it.
- Effector: Receives signals from the control center and acts to bring the variable back to the set point. Effectors might be muscles, glands, or organs that can adjust their functions to restore balance.



- Feedback Mechanism:

- Negative Feedback: The most common form, where the response reverses the change in the variable, bringing it back to the set point.
- Positive Feedback: Less common, amplifying the change instead of reversing it (e.g., blood clotting, childbirth).

RECEPTOR

- Thermoreceptors: Found in the skin and hypothalamus, these receptors detect changes in body temperature.
- Baroreceptors: Located in blood vessels, they detect changes in blood pressure.
- Chemoreceptors: Found in the carotid bodies and aortic arch, these sensors detect changes in blood oxygen, carbon dioxide, and pH levels.

CONTROL CENTER

- Hypothalamus: Acts as a controller for body temperature, thirst, hunger, and other homeostatic functions.
- Pancreas: Functions as a controller for blood glucose regulation. It releases insulin or glucagon based on blood sugar levels detected by sensors.
- Brainstem (Medulla): Regulates vital functions like heart rate, respiratory rate, and blood pressure by processing signals from baroreceptors and chemoreceptors.

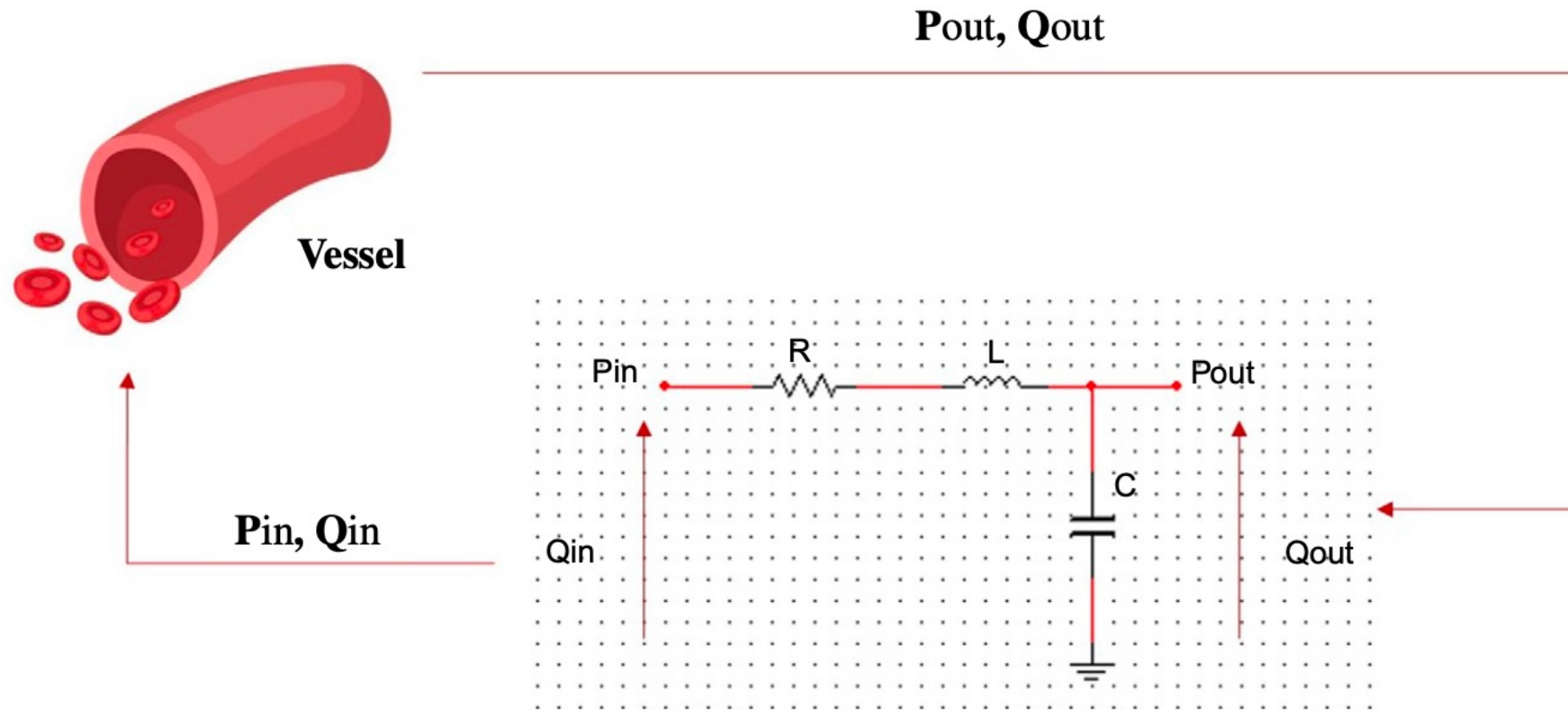
EFFECTOR

- Muscles: In thermoregulation, skeletal muscles act as effectors by shivering to generate heat when body temperature drops below the set point.
- Glands: The pancreas acts as an effector by releasing insulin when blood glucose levels are high or glucagon when they're low, thereby regulating blood sugar levels.
- Organs: In the regulation of blood pressure, the heart and blood vessels serve as effectors. The heart may beat faster or slower, and blood vessels can constrict or dilate in response to signals from the control center.

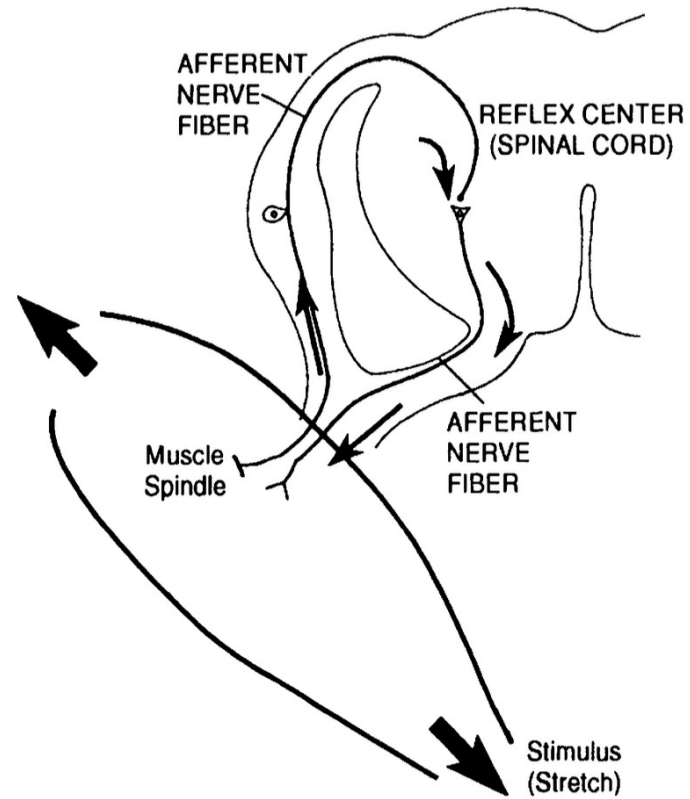
CARDIOVASCULAR SYSTEM

- Thermodynamics of vessels was represented by an electronic circuit
- voltage represents the blood pressure
- current represents the blood flow
- C and R are the compliance and the resistance of the systemic arterial tree, respectively.
- L is the impedance of proximal aorta.

EQUIVALENT ELECTRONIC CIRCUIT OF BLOOD VESSELS SYSTEM



MUSCLE STRETCH REFLEX

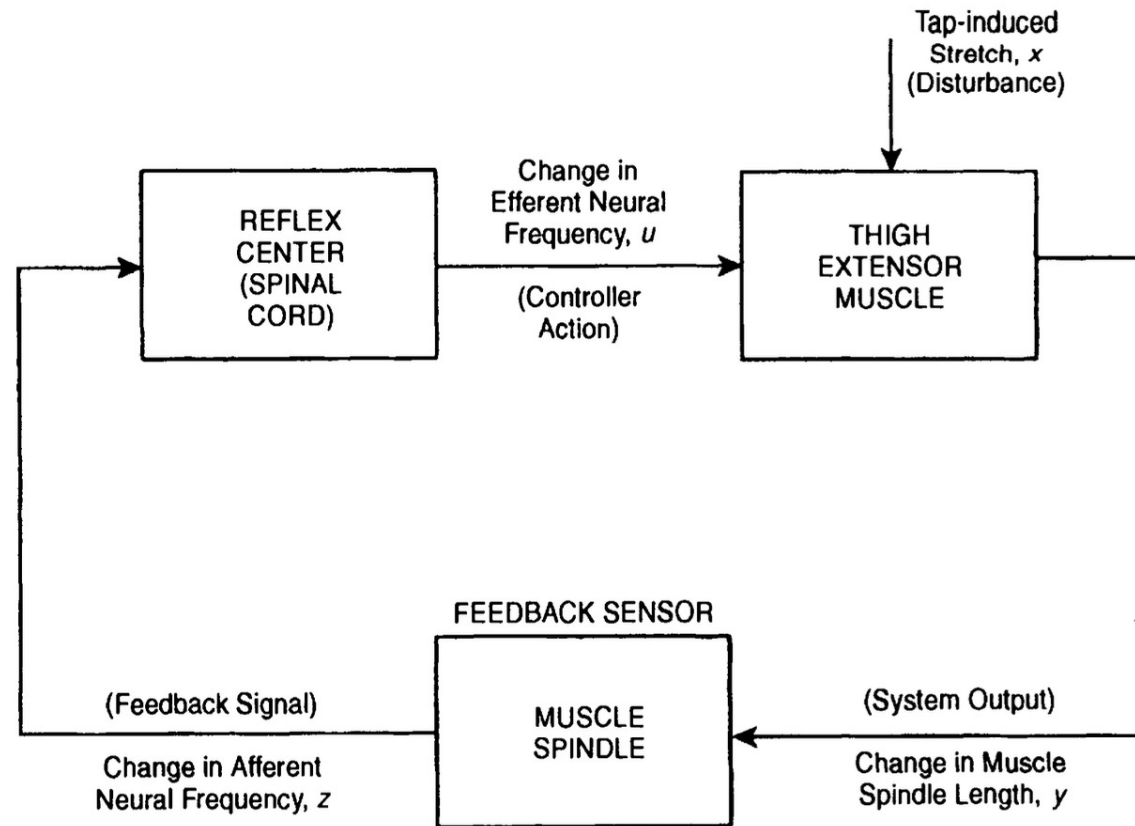


- Kneejerk: used in routine medical examinations as an assessment of the state of the nervous system.
- A sharp tap to the patellar tendon in the knee leads to an abrupt stretching of the extensor muscle in the thigh to which the tendon is attached. This activates the muscle spindles, which are stretch receptors.
- Neural impulses, which encode information about the magnitude of the stretch, are sent along afferent nerve fibers to the spinal cord.

- Since each afferent nerve is synaptically connected with one motor-neuron in the spinal cord, the motor-neurons get activated and, in turn, send efferent neural impulses back to the same thigh muscle.
- These produce a contraction of the muscle, which acts to straighten the lower leg.

- monosynaptic: only two neurons and one synapse are employed in the reflex.
- reflex: closed-loop regulation of muscle length is accomplished in a completely involuntary fashion.
- feedback type: negative

BLOCK DIAGRAM REPRESENTATION

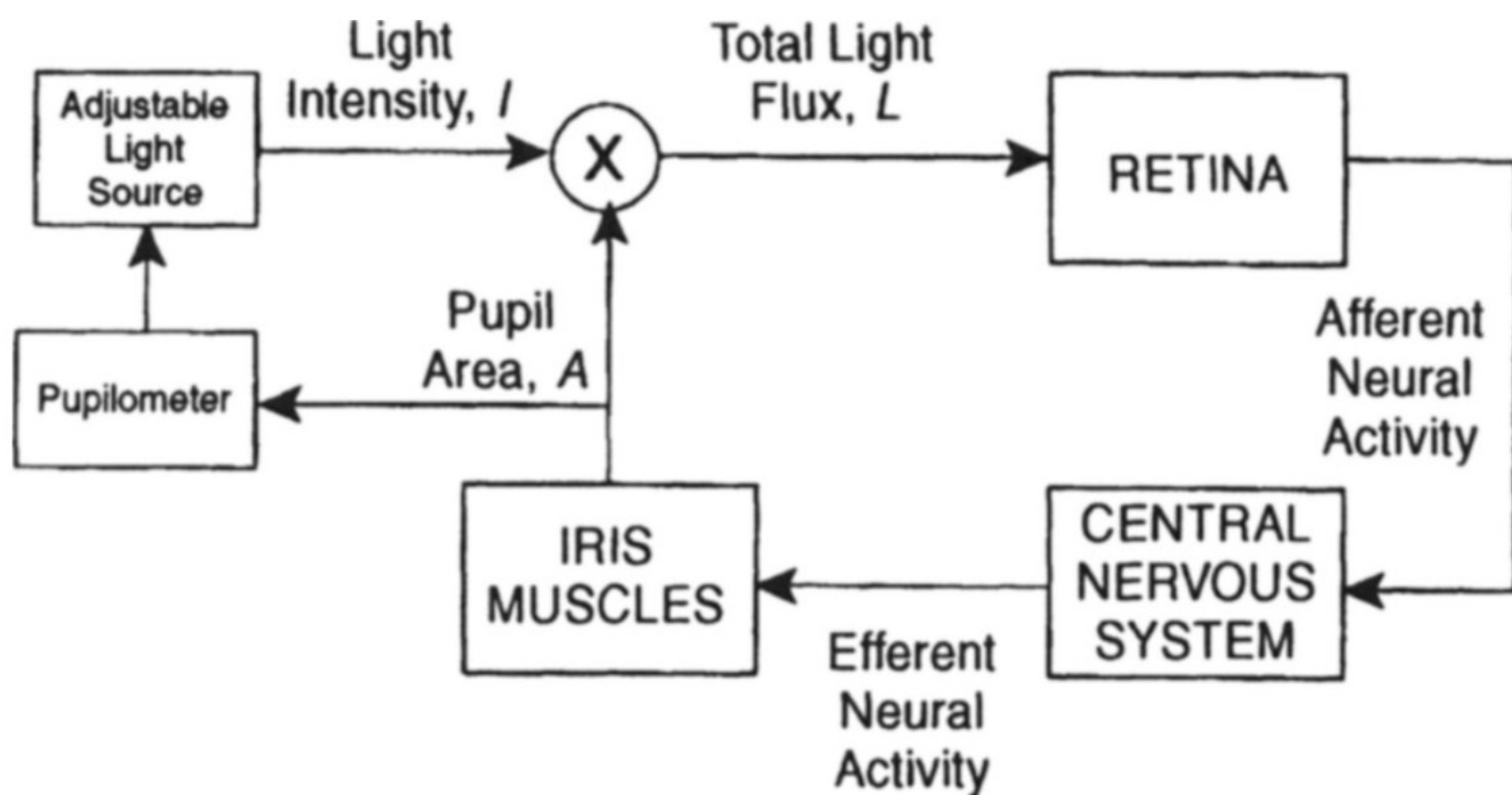


- The disturbance (x) is the amount of initial stretch produced by the tap to the knee.
- This produces a proportionate amount of stretch (y) in the muscle spindles, which act as the feedback sensor.
- The spindles translate this mechanical quantity into an increase in afferent neural traffic (z) sent back to the reflex center in the spinal cord, which corresponds to controller.

- In turn, the controller action is an increase in efferent neural traffic (u) directed back to the thigh muscle, which subsequently contracts in order to offset the initial stretch.
- negative feedback system: since the initial disturbance leads to a controller action that is aimed at reducing the effect of the disturbance.

PUPILLARY LIGHT REFLEX

- In response to a decrease in light intensity, receptors in the retina transmit neural impulses at a higher rate to the pretectal nuclei in the midbrain, and subsequently, to the Edinger- Westphal nuclei.
- From the Edinger-Westphal nuclei, a change in neural traffic down the efferent nerves back to the eyes leads to a relaxation of the sphincter muscles and contraction of the radial dilator muscles that together produce an increase in pupil area, which increases the total flux of light falling on the retina.



BAINBRIDGE REFLEX

- Cardiac reflex that aids in the matching of cardiac output (the flow rate at which blood is pumped out of the heart) to venous return (the flow rate at which blood returns to the heart).
- Suppose there is a transient increase in the amount of venous blood returning to the right atrium.
- This increases blood pressure in the right atrium, stimulating the atrial stretch receptors.
- As a result, neural traffic in the vagal afferents to the medulla is increased.

- This, in turn, leads to an increase in efferent activity in the cardiac sympathetic nerves as well as a parallel decrease in efferent parasympathetic activity.
- Consequently, both heart rate and cardiac contractility are increased, raising cardiac output.
- In this way, the reflex acts like a servomechanism, adjusting cardiac output to track venous return.

BARORECEPTOR REFLEX

- Suppose arterial blood pressure falls. This reduces the stimulation of the baroreceptors located in the aortic arch and the carotid sinus, which lowers the rate at which neural impulses are sent along the glossopharyngeal and vagal afferents to the autonomic centers in the medulla.
- Consequently, sympathetic neural outflow is increased, leading to an increase in heart rate and cardiac contractility, as well as vasoconstriction of the peripheral vascular system.
- At the same time, a decreased parasympathetic outflow aids in the heart rate increase. These factors together act to raise arterial pressure.

WATER BALANCE

- The control system is intimately coupled with the control of sodium excretion. When sodium is reabsorbed by the distal tubules of the kidneys, water will also be reabsorbed if the permeability of the tubular epithelium is lowered.
- When there is a drop in plasma volume, mean systemic pressure decreases, leading to a change in stimulation of the left atrial pressure receptors.

- The latter send signals to a group of neurons in the hypothalamus, increasing its production of vasopressin or antidiuretic hormone (ADH).
- As a result, the ADH concentration in blood plasma increases, which leads to an increase in water permeability of the kidney distal tubules and collecting ducts.

COMPARISON

- An engineering control system is designed to accomplish a defined task, and frequently, the governing parameters have been fine-tuned extensively so that the system will perform its task in an "optimal" manner.
- In contrast, physiological control systems are built for versatility and may be capable of serving several different functions. For instance, although the primary purpose of the respiratory system is to provide gas exchange, a secondary but also important function is to facilitate the elimination of heat from the body.

- Since the engineering control system is synthesized by the designer, the characteristics of its various components are generally known.
- On the other hand, the physiological control system usually consists of components that are unknown and difficult to analyze.

- There is an extensive degree of cross-coupling or interaction among different physiological control systems.
- The proper functioning of the cardiovascular system, for instance, is to a large extent dependent on interactions with the respiratory, renal, endocrine, and other organ systems.

- Physiological control systems, in general, are adaptive.
- System may be able to offset any change in output not only through feedback but also by allowing the controller or plant characteristics to change.
- For instance, muscle stretch reflex plays a protective role in regulating muscle stretch, it also can hinder the effects of voluntary control of the muscles involved.

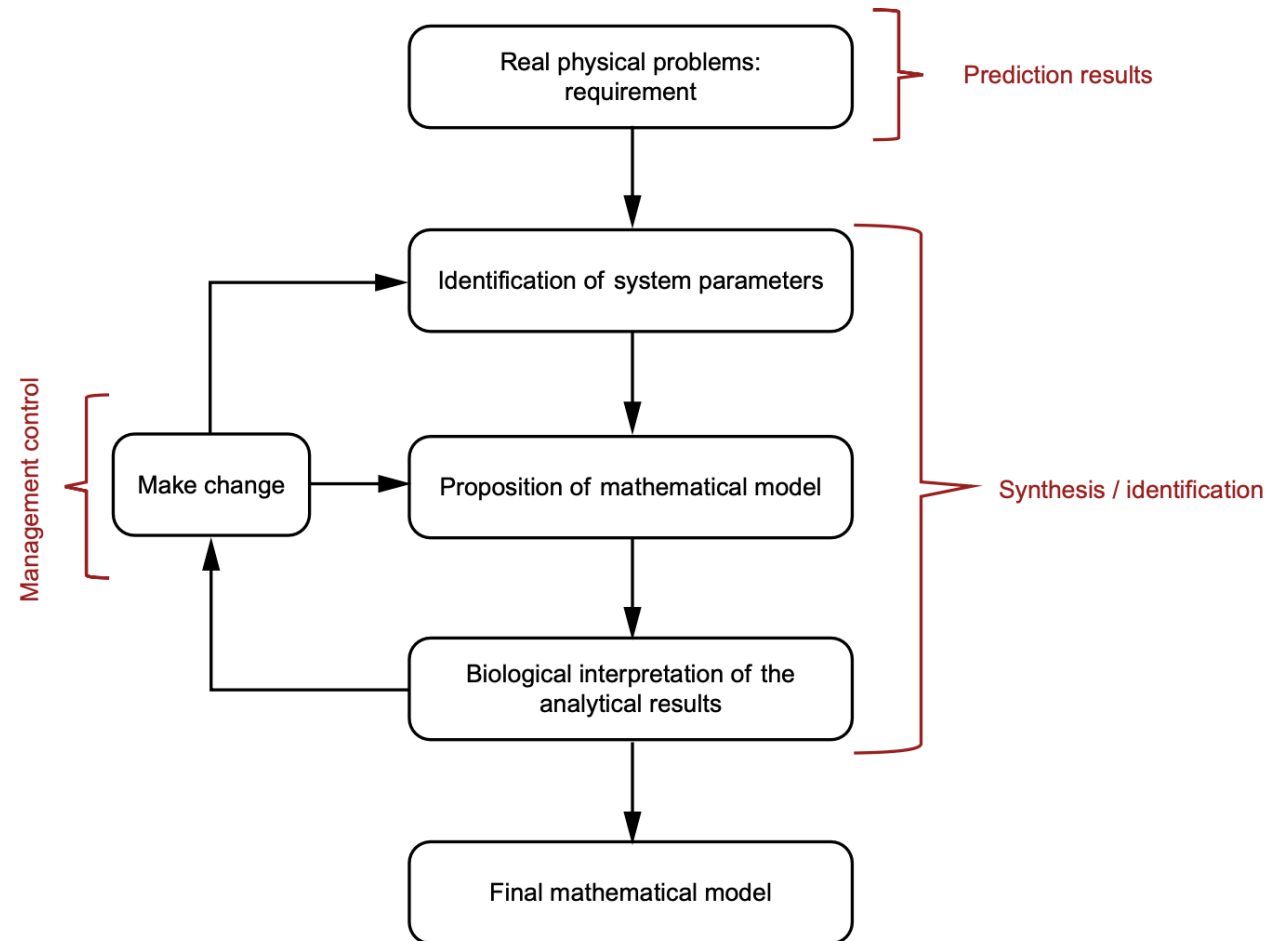
- In engineering control systems, the feedback signal is explicitly subtracted from the reference input, demonstrating clearly the use of negative feedback.
- However, negative feedback in the stretch reflex system is "built into" the plant characteristics: increased efferent neural input produces a contraction of the extensor muscle, thereby acting to counteract the initial stretch. This kind of embedded feedback is highly common in physiological systems.

- Physiological systems are generally nonlinear, while engineering control systems can be linear or nonlinear.

THE ART OF MODELING

- Physiological modeling helps to understand how the body's components interact, and how these interactions change in disease or after therapy.
- It can also help to predict how a system will behave in an altered state.
- Block diagrams is useful to formalize belief (which is usually based observations and partly on intuition) about how the various processes involved are causally related.
- The block diagram represents a conceptual model of the physiological control system under study.

- However, such a model is limited in its ability to enhance understanding or make predictions, since it only allows qualitative inferences to be made.
- To advance the analysis to the next level involves the upgrading of the conceptual model into a mathematical model.
- The mathematical model allows us to make hypotheses about the contents in each of the "boxes" of the block diagram.



- The power of the modeling process lies in its replication of the scientific method: the discrepancy between model prediction and physiological observation can be used as "feed-back" to alert the inadequacies of one or more component hypotheses.
- This allows to return to the model development stage once again in order to modify the previous assumptions.

- On the basis of the particular physical or chemical laws involved, an algebraic, differential, or integral equation are derived that relates the input to the output of the system component.
- The inherent self-consistency of the model derives from the operational rules of mathematics, which provide a logical accounting system for dealing with the multiple system variables and their interactions with one another.

- Subsequently, retest the revised model against experimental observations, and so the alternating process of induction and deduction continues until the model can explain most of the observed behavior.
- Then, having arrived at a good model, use this model to predict how the system might behave under experimental conditions that have not been employed previously.
- These predictions would serve as a guide for the planning and design of future experiments.

PARAMETRIC MODELS

- use a predefined mathematical form (such as linear equations, transfer functions, or differential equations) that includes a set of parameters that describe the system.
- parameters are estimated from data and represent specific physiological aspects of the system.
- E.g.: linear equations, differential equations and transfer functions

- Transfer Function Models: used to describe the relationship between input and output signals of a system, often in the frequency domain.
- E.g.: a transfer function relates blood pressure (input) to blood flow (output) in cardiovascular modeling.

- Compartmental Models: a physiological system is divided into compartments (E.g.: blood, tissue, organs), each with its own parameters (E.g.: rate constants).
- E.g.: pharmacokinetics describes how a drug moves through the body.

- State-Space Models: represent systems with a set of variables (states) that describe the system's status over time. In physiological modeling, a state-space approach could model
- E.g.: glucose levels in the body, where states like insulin and glucose levels are connected through equations.

- Ordinary Differential Equation (ODE) Models: represent dynamic systems by describing how system variables change over time.
- E.g.: hormone concentration changes in response to feedback control mechanisms.

- Glucose-Insulin Regulation Model: represents insulin secretion and glucose uptake. Parameters: insulin sensitivity and pancreatic response rate.
- Cardiovascular System Models: describe the dynamics of heart rate, blood pressure, and blood flow. Parameters: vascular resistance, compliance of blood vessels, and heart contractility.

- Pharmacokinetic Models: describe drug absorption, distribution, metabolism, and elimination. Parameters: absorption rates, clearance rates, and volume of distribution.

NON-PARAMETRIC MODELS

- Directly use data to characterize the system's behavior without making assumptions about the underlying parameters.
- E.g.: Machine learning, impulse response analysis

- Machine Learning Models: can learn complex patterns in large datasets without needing specific assumptions about the underlying physiology.
- useful in tasks like predicting disease outcomes or physiological responses based on input variables.
- E.g., patient history, biomarkers.

- Kernel-Based Methods: useful for non-linear regression and classification.
- They can model physiological signals, like EMG or ECG patterns, by mapping data into higher-dimensional spaces to find patterns not evident in the original space.

- Fourier and Frequency Analysis: used in signal processing to decompose signals (e.g., heart rate variability, EEG) into frequencies.
- identifies dominant frequencies or oscillatory behavior without fitting data to a parametric model.

- Impulse and Step Response Models: capture a system's response to an input signal without requiring specific parameters.
- E.g.: impulse response analysis in cardiovascular physiology might involve observing the heart rate's response to a small perturbation in blood pressure.

- Heart Rate Variability Analysis: spectral analysis (Fourier transform) allow for decomposing the heart rate signal into different frequency components.
- helps in identifying stress levels, autonomic regulation, and other health indicators without needing assumptions about the underlying system.

- EMG Signal Processing in Neuromuscular Control: neural networks can be applied to electromyography (EMG) data to predict muscle activation patterns or motor intentions.
- Since EMG signals are complex and influenced by various physiological and environmental factors, machine learning models can effectively capture these nonlinear dynamics.

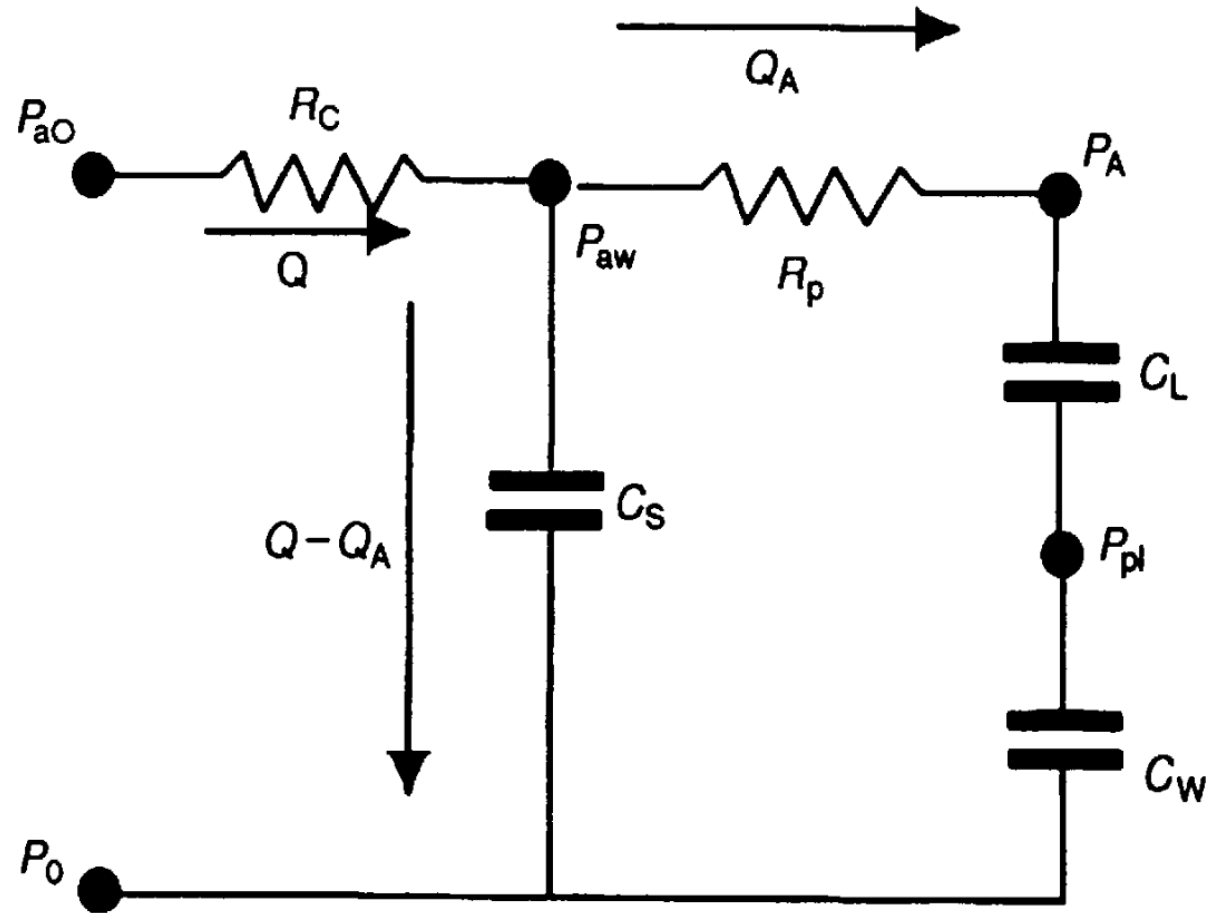
- Respiratory Signal Analysis: Non-parametric models can analyze respiratory signals to study patterns in breathing or detect abnormalities.
- kernel methods can be used to classify respiratory sounds, like wheezing or coughing, without needing specific model assumptions about airflow dynamics.

- Predictive Health Models: Machine learning models are commonly used to predict disease onset or outcomes based on patient data.
- neural network might use historical blood glucose and insulin data to predict the likelihood of hyperglycemia in diabetic patients.

LUMPED PARAMETER SYSTEMS

- System is divided into discrete parts or compartments, each with uniform properties.
- Parameters such as resistance, capacitance, or volume are assumed to be concentrated at specific points or in specific "lumps."
- E.g.: blood flow can be modeled as a single compartment with an overall resistance and compliance, ignoring variations along the blood vessels

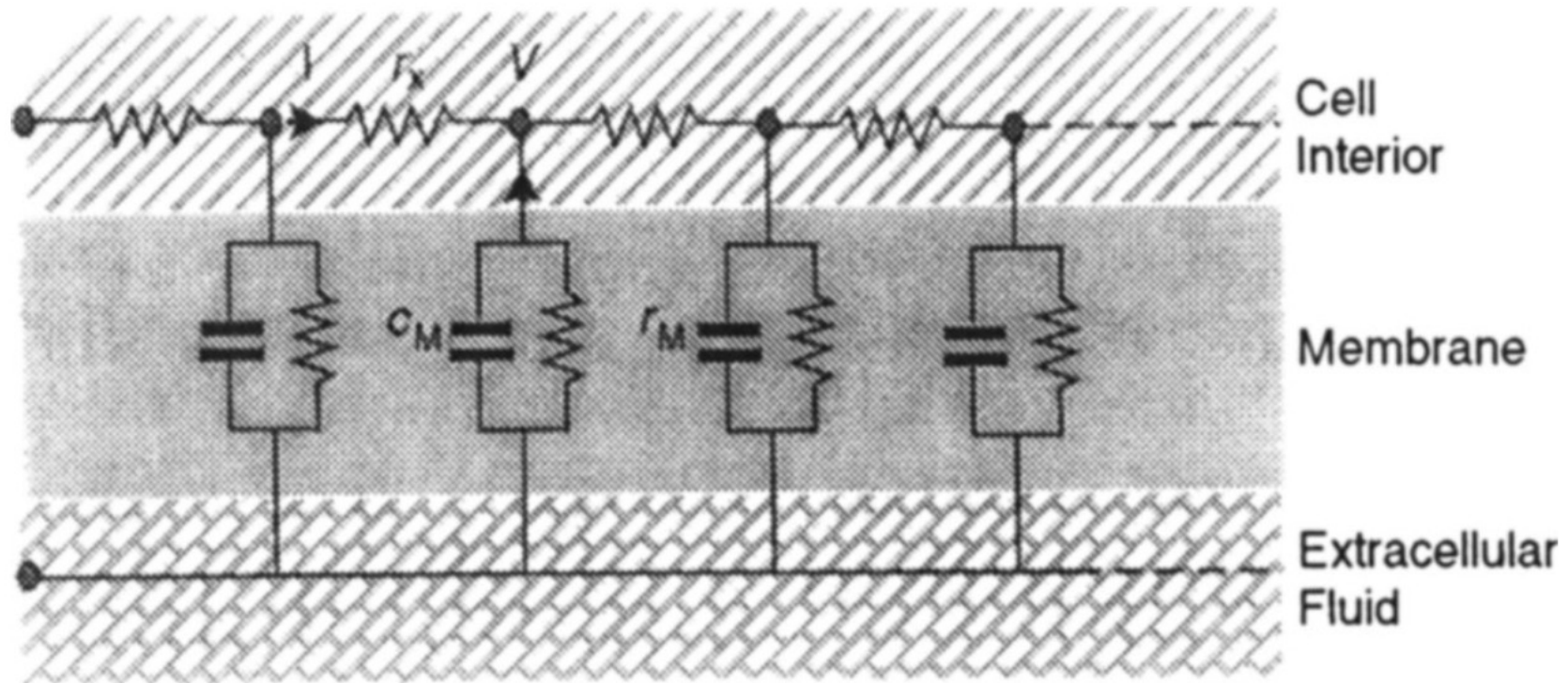
RESPIRATORY MECHANICS



DISTRIBUTED PARAMETER SYSTEMS

- spatial variations within a system.
- Parameters are spread out or "distributed" continuously over the length, area, or volume of the system.
- E.g.: modeling heat transfer across skin layers, or the diffusion of ions in nerve axons.

UNMYELINATED NERVE FIBER



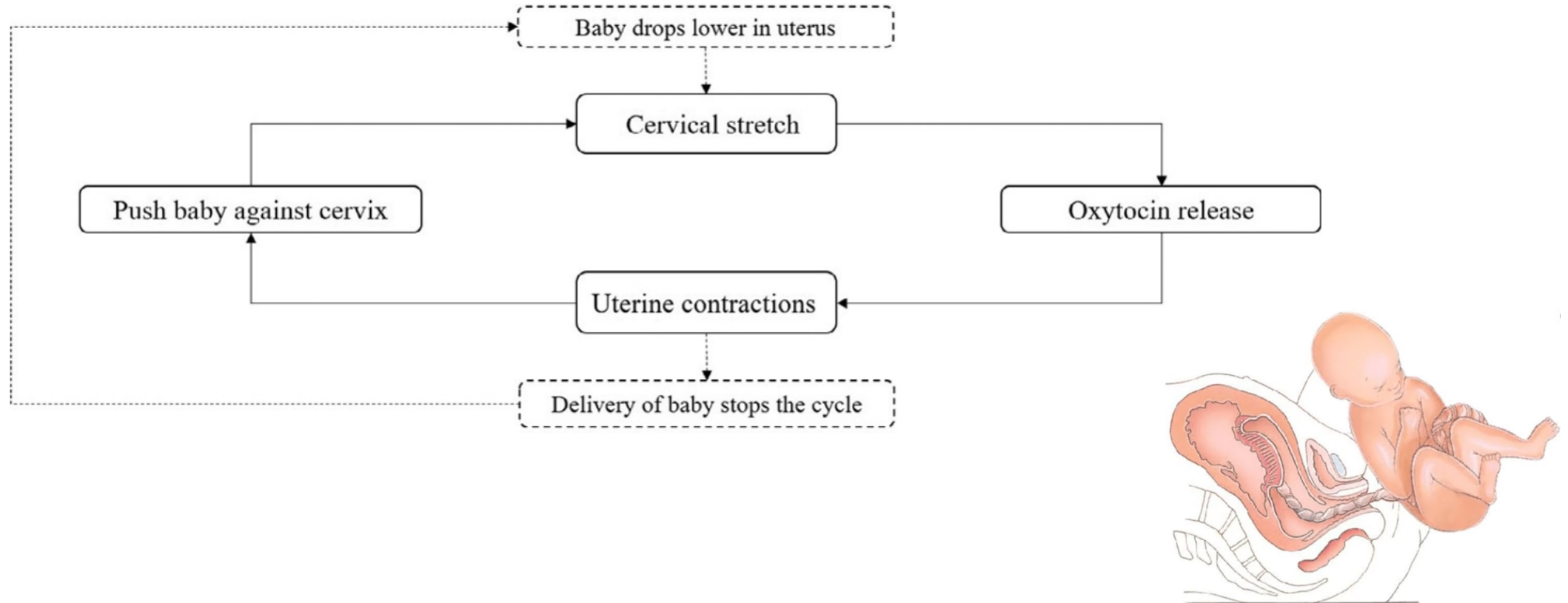
FEEDBACK MECHANISM

- A feedback mechanism in a physiological control system is a process that helps regulate and maintain homeostasis by responding to changes in a controlled variable.
- Negative Feedback: maintains stability by keeping physiological variables within a healthy range.
- Positive Feedback: promotes rapid changes needed in specific situations and completes a function effectively.

POSITIVE FEEDBACK

- Childbirth: During labor, the hormone oxytocin is released, which intensifies uterine contractions. These contractions push the baby toward the cervix, stretching it further, which signals the release of more oxytocin. This cycle continues until birth occurs, completing the process.
- Blood Clotting: When a blood vessel is damaged, platelets stick to the injury site and release chemicals that attract more platelets. This loop continues until a clot is formed, effectively sealing the wound.
- Lactation: Suckling by an infant stimulates the release of oxytocin, which causes milk ejection. The more the baby suckles, the more milk is ejected, until the infant stops feeding.

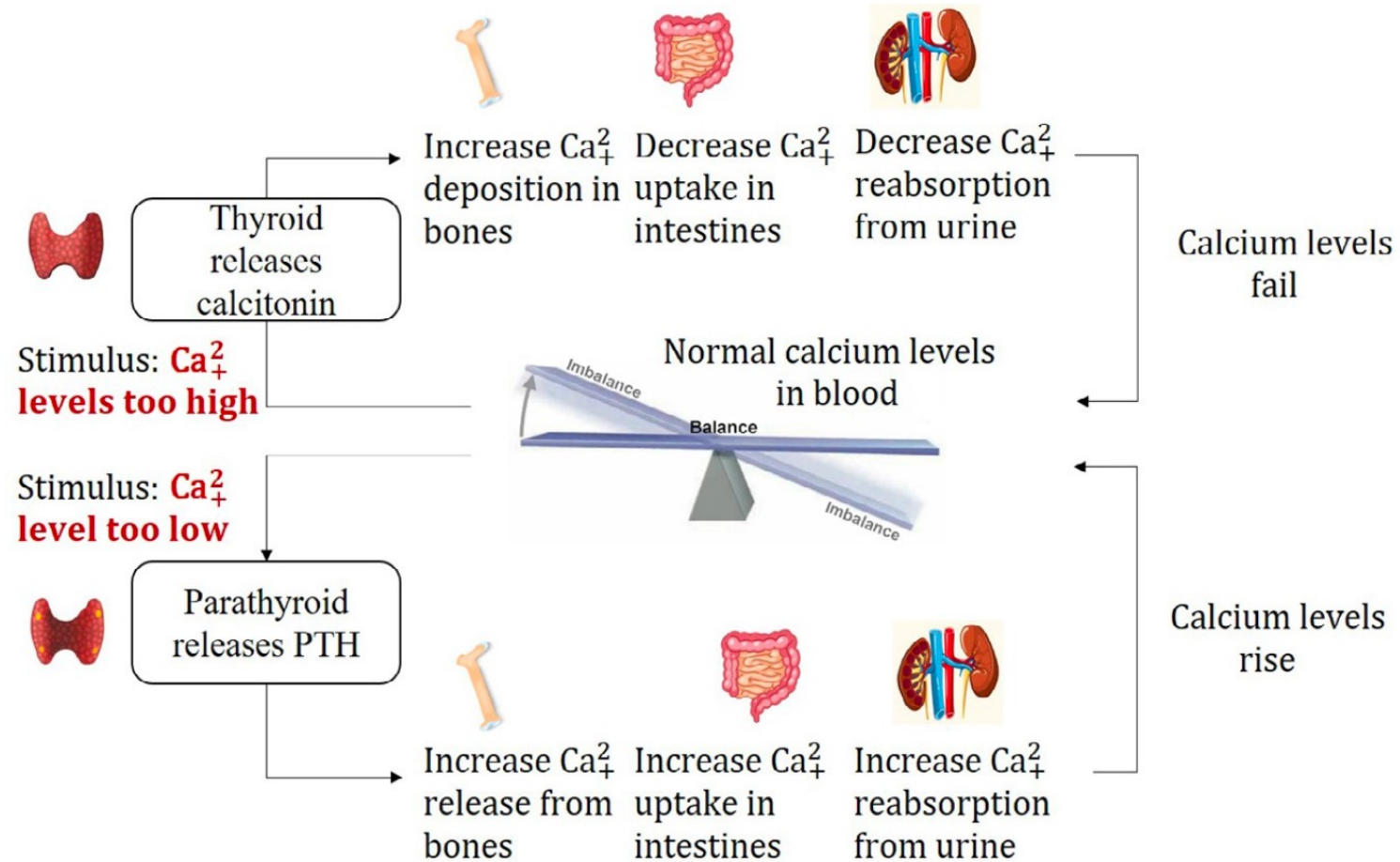
CHILDBIRTH



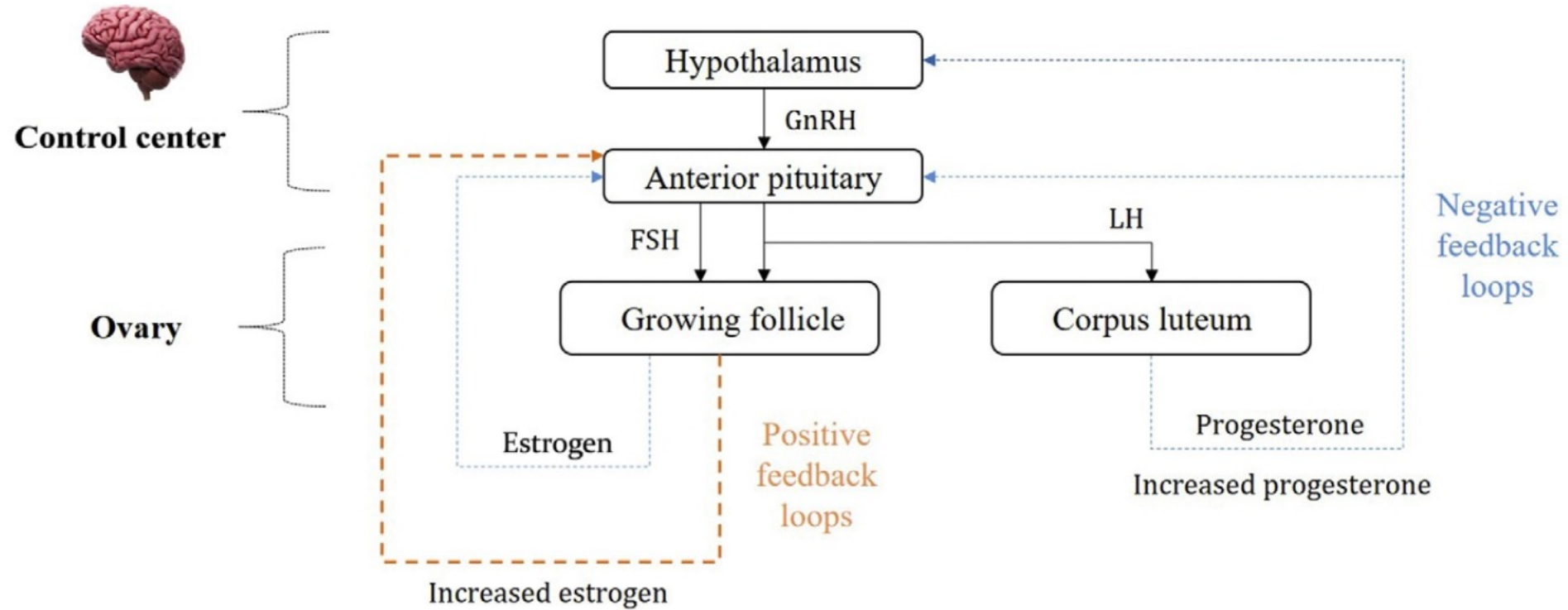
NEGATIVE FEEDBACK

- **Thermoregulation:** When body temperature rises above the set point, thermoreceptors in the skin and hypothalamus send signals to initiate cooling responses, such as sweating and vasodilation (expanding blood vessels near the skin). Conversely, if the temperature drops, responses like shivering and vasoconstriction (narrowing blood vessels) help increase body heat.
- **Blood Glucose Regulation:** When blood glucose levels rise after a meal, the pancreas releases insulin, which signals cells to absorb glucose, lowering blood sugar levels back to the set point. If glucose levels are too low, the pancreas releases glucagon to increase blood glucose levels.
- **Blood Pressure Control:** When blood pressure rises, baroreceptors (pressure-sensitive cells) in the arteries send signals to the brain to lower the heart rate and dilate blood vessels, decreasing blood pressure. If blood pressure is too low, the opposite response is triggered.

BLOOD CALCIUM REGULATION



POSITIVE AND NEGATIVE FEEDBACK LOOPS: MENSTRUAL CYCLE



UNIT - V - PROPERTIES OF PHYSIOLOGICAL CONTROL SYSTEMS

Introduction Generalized system properties - Adaptive and constrained systems - Models with combination of system elements - Linear model of respiratory mechanics –Linear model of lung mechanics

C05: Summarize the properties of physiological control systems
(Understand)

GENERALIZED SYSTEM PROPERTIES

- Resistive property (R)
- Storage property (C)
- Inertance property (L)

RESISTIVE PROPERTY

$$V = IR$$

- where V is the voltage or driving potential across the resistor and I represents the current that flows through it.
- V is an "across" variable and may be viewed as a measure of "effort".
- I is a "through" variable and represents a measure of "flow".

GENERALIZED SYSTEM

- if we define the generalized "effort" variable, Ψ , and the generalized "flow" variable, ζ , Ohm's law becomes:

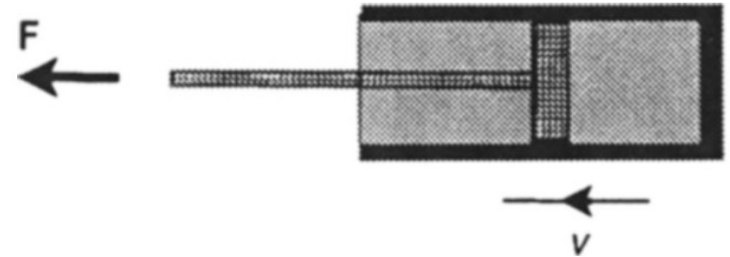
$$\psi = \zeta R$$

- where R now represents a generalized resistance.

MECHANICAL SYSTEM

- when a force F is applied to the plunger, it will move with a velocity v that is proportional to F .

$$F = vR_m$$

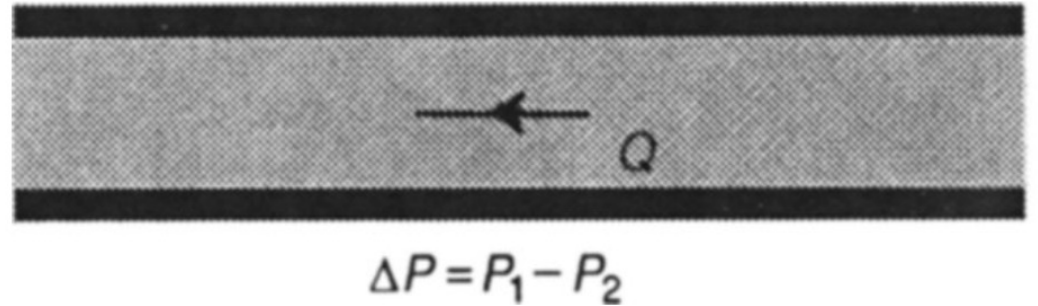


- constant of proportionality, R_m , which is related to the viscosity of the fluid inside the dashpot, provides a measure of "mechanical resistance".

FLUIDIC SYSTEM

- Poiseuille's law: the volumetric flow of fluid (Q) through a rigid tube is proportional to the pressure difference (ΔP) across the two ends of the tube.

$$\Delta P = QR_f$$

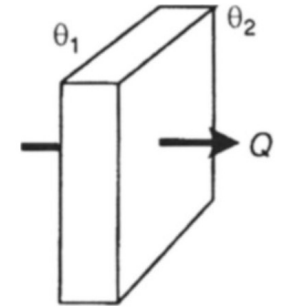


- fluid resistance, R_f , is directly related to the viscosity of the fluid and the length of the tube, and inversely proportional to the square of the tube cross-sectional area.

THERMAL SYSTEM

- Fourier's law of thermal transfer: the flow of heat, Q , conducted through a given material is directly proportional to the temperature difference that exists across the material.

$$\Delta\theta = QR_t$$

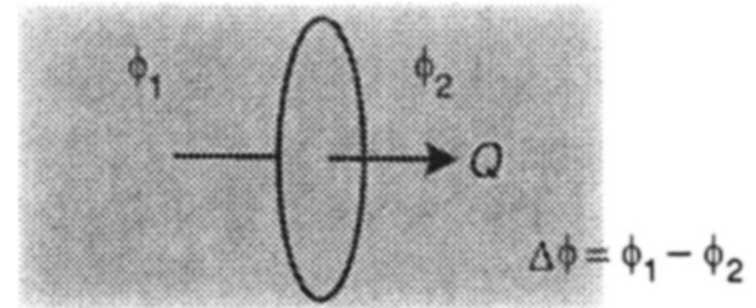


- Thermal resistance, R_t can be shown to be inversely related to the thermal conductivity of the material.

CHEMICAL SYSTEM

- Fick's law of diffusion: the flux, Q , of a given chemical species across a permeable membrane separating two fluids with different species concentrations is proportional to the concentration difference, $\Delta \phi$.

$$\Delta \phi = QR_c$$



- diffusion resistance, R_c , is inversely proportional to the more commonly used parameter, the membrane diffusivity.

- resistive property also represent elements through which energy is dissipated.
- in electrical systems, the product of voltage and current yields power.
- in mechanical systems, force multiplied by velocity also yields power.
- in fluidic systems, power is defined as the product of pressure and flow rate.
- Thus, in any general system, energy is dissipated when effort is applied to produce flow through the resistive element.

STORAGE PROPERTY

- In electrical systems, this takes the form of capacitance, defined as the amount of electrical charge (q) stored in the capacitor per unit voltage (V) that exists across the capacitor: $C = \frac{q}{V}$

- q represents the accumulation of all electric charge delivered via current flow to the capacitor, so that the following relationship exists between q and I : $q = \int_0^t I dt$

GENERALIZED SYSTEM

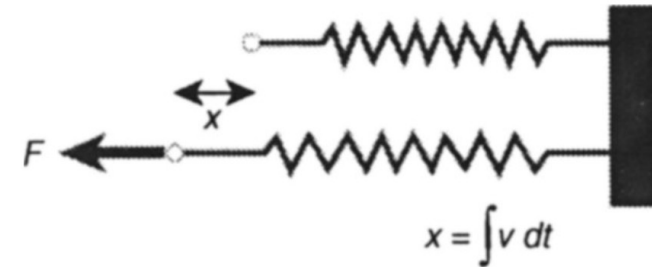
$$\psi = \zeta R$$

$$\psi = \frac{1}{c} \int_0^t \zeta \, dt$$

MECHANICAL SYSTEM

- In elastic spring, for a given applied force, the mechanical compliance determines the extent to which the spring will be extended or compressed.

$$x = FC_m$$

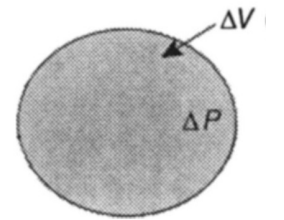


- This property is also inversely related to the stiffness or elastic modulus of the spring: the more compliant the spring, the less stiff it will be, and the more it will extend for a given applied tension.

FLUIDIC SYSTEM

- in fluid-filled balloon, compliance determines the volume by which the balloon will expand or contract per unit change in applied pressure.
- compliance here is determined primarily by the elasticity of the balloon material: the stiffer the material, the less "compliant" the balloon.

$$\Delta V = \Delta P C_f$$

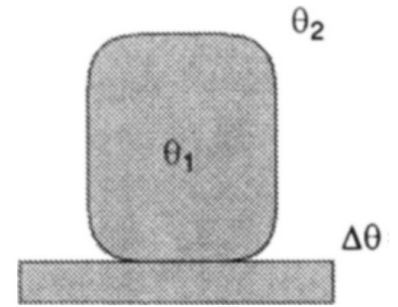


- a much smaller contribution to the compliance arises from the compressibility of the fluid inside the balloon.

THERMAL SYSTEM

- the "thermal mass" represents the amount of heat stored in a certain medium per unit difference in temperature that exists between the interior and exterior.

$$q = \Delta\theta C_t$$



- thermal capacitance depends on the dimensions and specific heat of the medium.

CHEMICAL SYSTEM

- in chemical systems, the storage property is represented by the total volume of the fluid in which the chemical species exists.
- for given volume, the total mass of the chemical species present is proportional to its concentration.

- storage property also represent elements through which energy is stored.
- the capacitor allows the accumulation of "static" or potential energy.
- in the mechanical spring, application of force produces an extension of length will lead to the storage of potential energy.
- in the balloon system, potential energy is stored when the balloon is expanded with the application of internal pressure.

INERTANCE PROPERTY

- allows the storage of kinetic energy in electrical systems.
- also known as inductance (L), which is defined as the voltage required to produce a given rate of change of electrical current.

$$V = L \frac{dI}{dt}$$

GENERALIZED SYSTEM

$$\psi = \zeta R$$

$$\psi = \frac{1}{C} \int_0^t \zeta \, dt$$

$$\psi = L \frac{d\zeta}{dt}$$

MECHANICAL SYSTEM

- Newton's Second Law of Motion: force equals mass times acceleration.
- In mechanical systems, ψ becomes force and ζ becomes velocity, and the inertance is simply the mass of the system.

FLUIDIC SYSTEM

- fluid acceleration is proportional to the pressure differential applied.

THERMAL AND CHEMICAL SYSTEM

- there is no element that represents inertance in thermal and chemical systems; kinetic energy storage does not exist in these systems.

CONSTRAINED SYSTEMS

- Control mechanisms that operate within predefined boundaries or limits to ensure stable, essential conditions within the body.
- designed to maintain consistency, often without significant adjustment to fluctuating conditions.
- These rigid controls are essential for keeping fundamental physiological parameters within narrow, life-sustaining ranges.

- systems typically have setpoints (target values or narrow ranges) that the system strictly regulates. These setpoints do not easily shift in response to external factors.
- less responsive to changes in the environment. Instead, they resist change to maintain stability, ensuring vital processes function correctly.
- E.g.: narrow range for blood pH levels (around 7.35-7.45) and body temperature (around 36.5-37.5°C).

- **Body Temperature Regulation:** Body temperature is maintained within a narrow range, around 36.5-37.5°C. Thermoregulatory mechanisms (like sweating, shivering, and blood flow adjustments) work to prevent deviations from this set range, especially in response to external temperature changes.
- **Blood Glucose Regulation:** Blood glucose levels are maintained within a relatively narrow range (typically 70-100 mg/dL when fasting) to ensure a stable supply of energy to tissues, especially the brain. Insulin and glucagon are released in precise amounts to maintain this balance, and deviations from this range can indicate diabetes or other metabolic disorders.

- **Blood pH Regulation:** Blood pH is tightly regulated between 7.35 and 7.45. The body employs multiple buffering systems (e.g., bicarbonate buffering) and organ systems (lungs and kidneys) to prevent blood pH from deviating outside this range.
- **Electrolyte Balance:** Electrolytes, like sodium and potassium, are regulated within narrow limits to maintain cellular functions, including nerve conduction and muscle contraction. Kidneys play a significant role in maintaining these levels through selective reabsorption and excretion, ensuring balance despite dietary intake variations.

- O_2 and CO_2 Levels: regulated through respiration to maintain a consistent supply of oxygen to tissues and to prevent acid-base imbalances. The respiratory system adjusts the breathing rate and depth to manage these gases within precise limits.
- Calcium Homeostasis: Blood calcium levels are closely regulated due to calcium's role in muscle contraction, nerve transmission, and blood clotting. The parathyroid hormone and vitamin D help keep calcium within the optimal range by regulating bone resorption, absorption from the gut, and kidney excretion.

ADAPTIVE SYSTEMS

- Control mechanisms that dynamically adjust in response to changes within the body or external environment.
- systems are characterized by their ability to learn from past experiences, refine responses over time, and operate with a high degree of flexibility.
- they are essential for maintaining homeostasis in conditions that are continually changing, such as during exercise, stress, or illness.

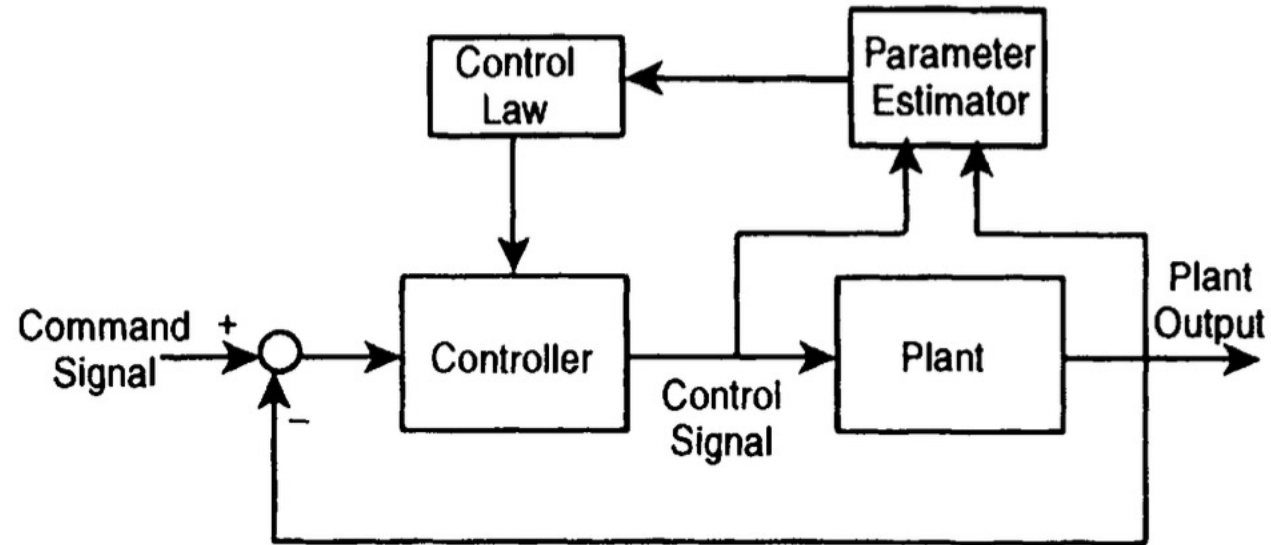
- Adaptive systems rely on feedback (typically negative feedback) to regulate physiological processes.
- This feedback allows the system to detect changes and make necessary adjustments.
- E.g.: heart rate variability in response to physical activity levels, respiratory rate adjustments based on oxygen and carbon dioxide levels in the blood.

- Cardiovascular System: During exercise, the cardiovascular system increases heart rate and blood pressure to supply muscles with oxygen. The heart's contractility and stroke volume can adapt to meet different levels of physical exertion, adjusting the body's baseline performance.
- Respiratory System: The respiratory rate increases with elevated CO₂ levels or during hypoxia (low oxygen) as sensed by chemoreceptors. The body adapts the breathing pattern to optimize oxygen intake and carbon dioxide expulsion.

- Immune System: When exposed to pathogens, it creates specific responses and retains "memory" cells for faster responses in future encounters with the same pathogen. Vaccination relies on this adaptability by training the immune system to recognize pathogens without exposure to the disease itself.
- Nervous System: Neuroplasticity allows the brain to reorganize and form new connections, especially after injury or in response to learning. The autonomic nervous system adapts to changing levels of stress, helping the body maintain a balance between alertness and relaxation.

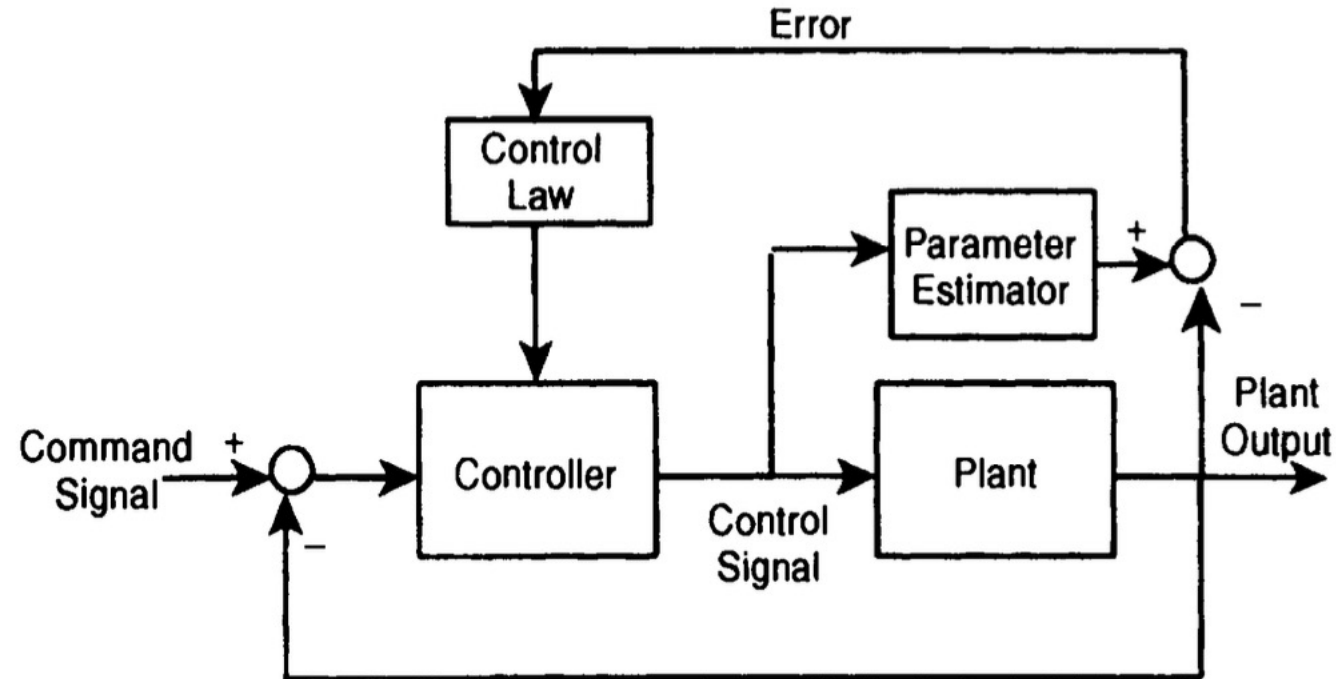
- Endocrine System: Hormonal regulation, such as the release of cortisol in response to stress, adjusts based on exposure and demand. Chronic stress can lead to adaptations in cortisol release patterns. The release of insulin in response to blood sugar levels also adapts in conditions like type 2 diabetes.

GENERAL ADAPTIVE CONTROL SYSTEM



- Parameter estimator: determine the changes in dynamics of the unknown plant.
- Control law: uses an optimization algorithm to select the control signal that is optimally adjusted for the altered plant dynamics.

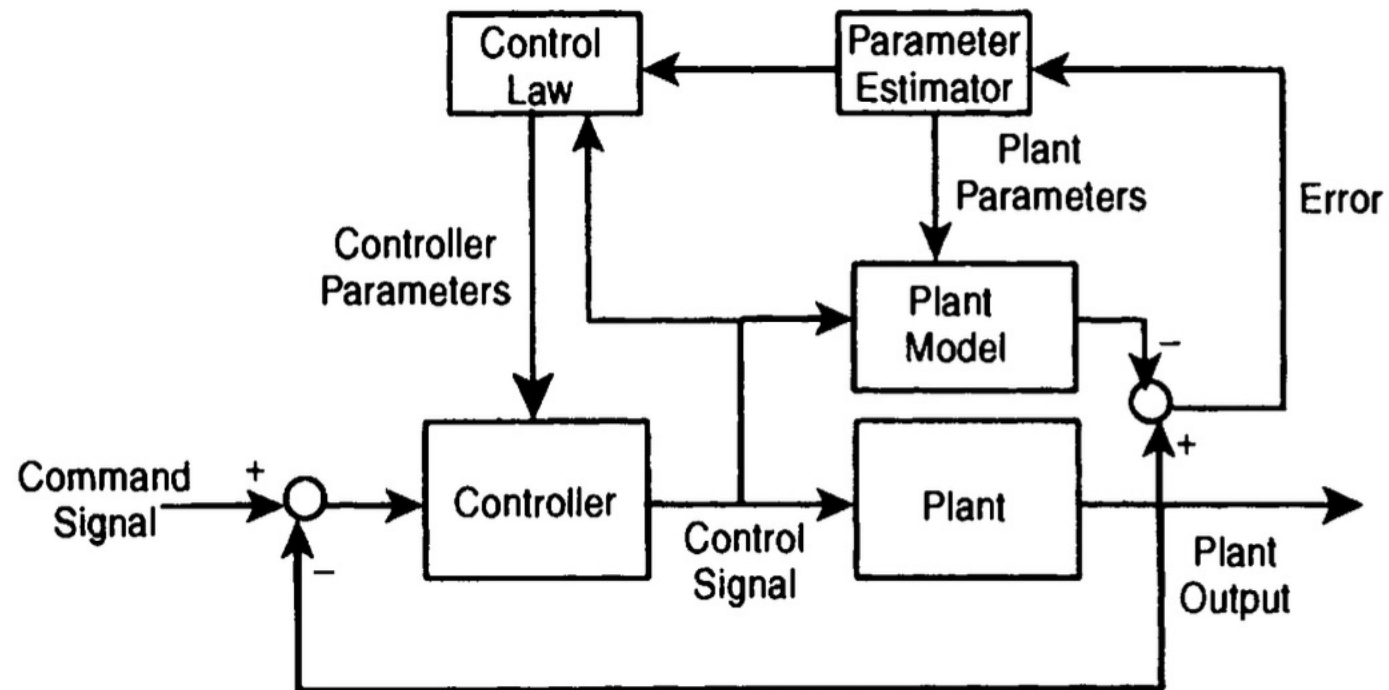
MULTIPLE-MODEL ADAPTIVE CONTROL SYSTEM



- A collection of models represents different possible system dynamics.
- The system continuously evaluates which model most accurately represents the current behavior of the plant (the controlled system).
- Based on the selected model, the control strategy is adapted to fit the specific dynamics.
- highly effective for abrupt changes in physiological states and is suited to situations where different models can represent distinct physiological conditions.

- Blood Glucose Regulation in Diabetes: used in some advanced artificial pancreas systems, where multiple models simulate different metabolic states (e.g., fasting, postprandial, exercise). By switching between models that represent different physiological states, MMAC can provide better blood glucose control by choosing the model closest to the patient's current metabolic condition.
- Cardiovascular Control: multiple models can represent different states of the cardiovascular system, such as rest, exercise, or stress, allowing the controller to switch as the patient's activity level or emotional state changes.

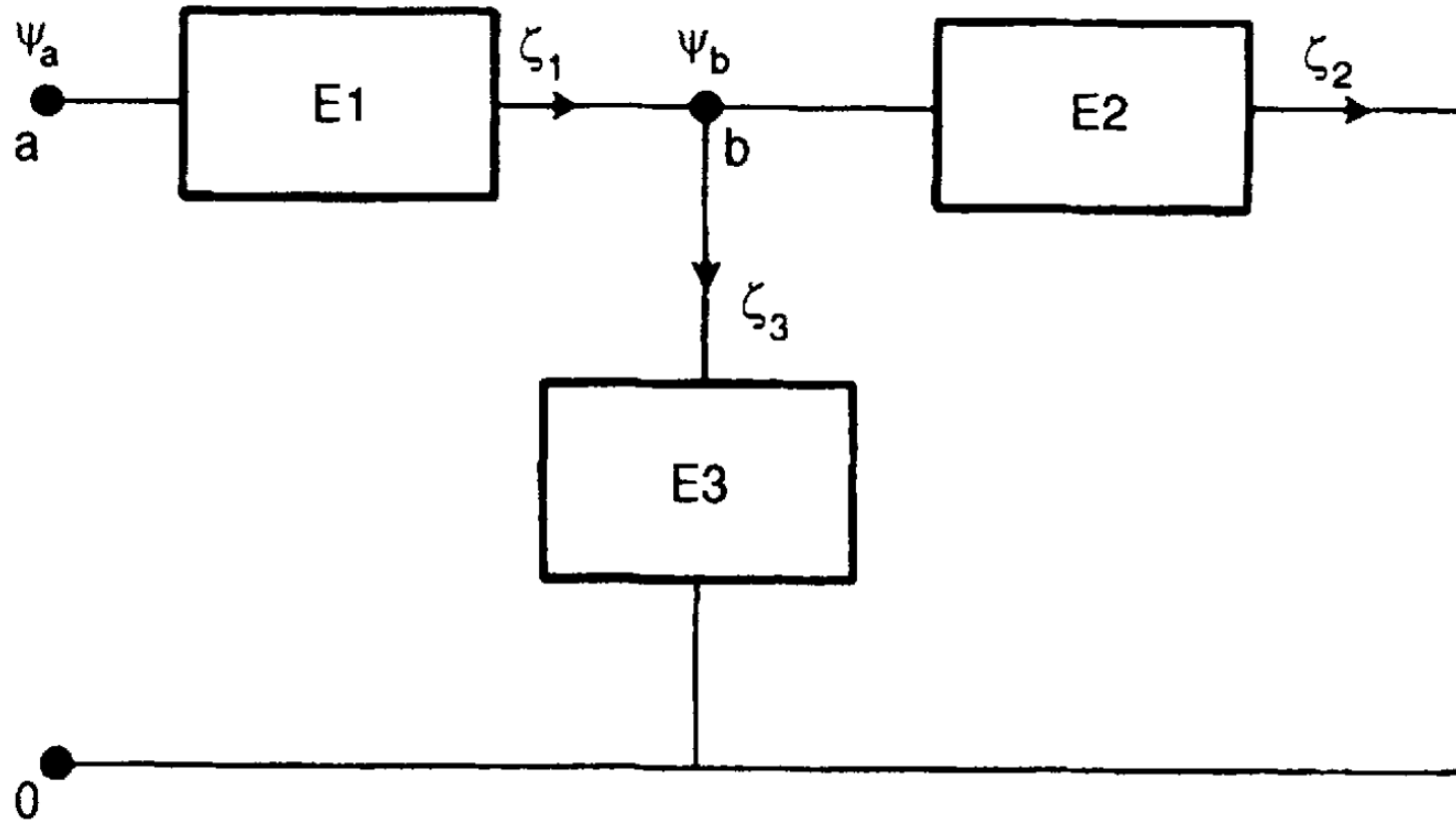
MODEL REFERENCE ADAPTIVE CONTROL SYSTEM



- Compares the behavior of the plant with a desired reference model that specifies ideal behavior.
- It adjusts control parameters so that the system output matches the reference model output as closely as possible.
- Effective in providing continuous control with gradual adjustments, ideal for applications that need steady performance relative to a reference model.

- Artificial Ventilation: used to ensure that ventilator settings adapt to mimic the reference model of normal breathing patterns, adjusting to the patient's changing lung compliance or airway resistance.
- Heart Rate Regulation: used to regulate heart rate in pacemakers by adjusting pacing parameters to match a reference model of ideal heart rhythm.

MODELS WITH COMBINATIONS OF SYSTEM ELEMENTS



- The algebraic sum of the "across-variable" values around any closed circuit must equal zero.
- in the loop a-b-0-a:

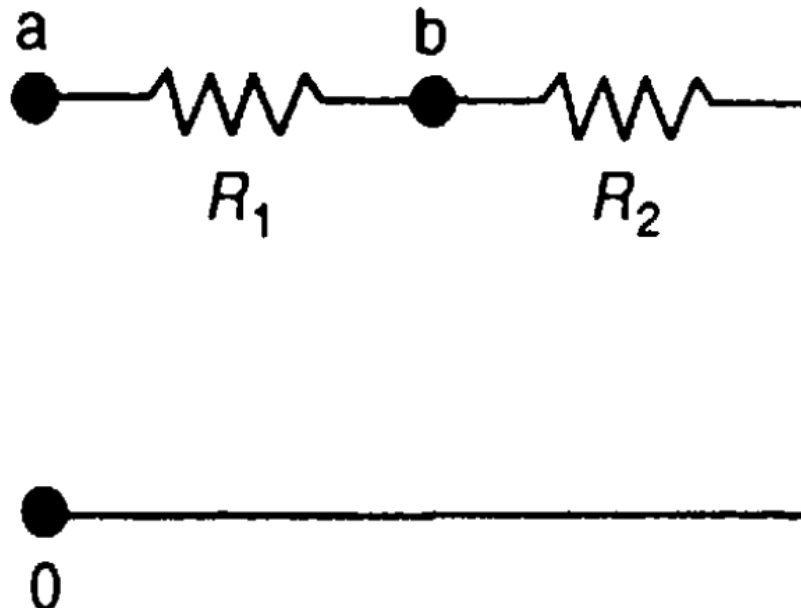
$$(\psi_a - \psi_b) + (\psi_b - \mathbf{0}) + (\mathbf{0} - \psi_a) = \mathbf{0}$$

- The algebraic sum of all "through-variable" values into a given node must equal zero.
- at node b:

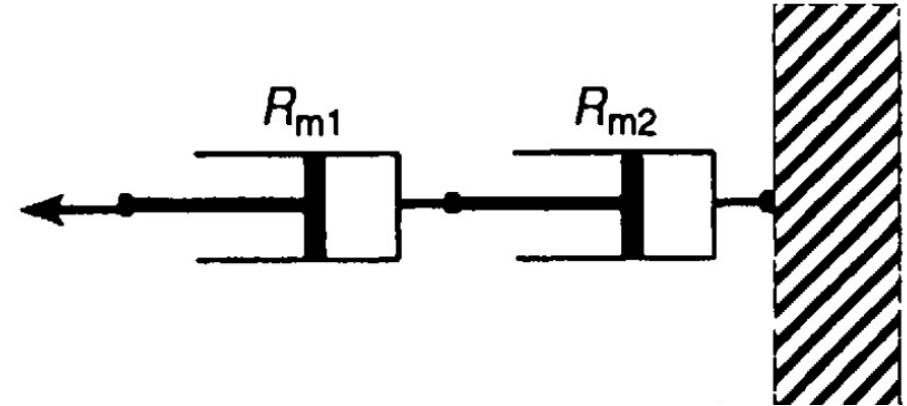
$$\zeta_1 = \zeta_2 + \zeta_3$$

$$\zeta_1 + (-\zeta_2) + (-\zeta_3) = 0$$

SERIES COMBINATIONS OF ELECTRICAL RESISTANCE AND MECHANICAL DASHPOT

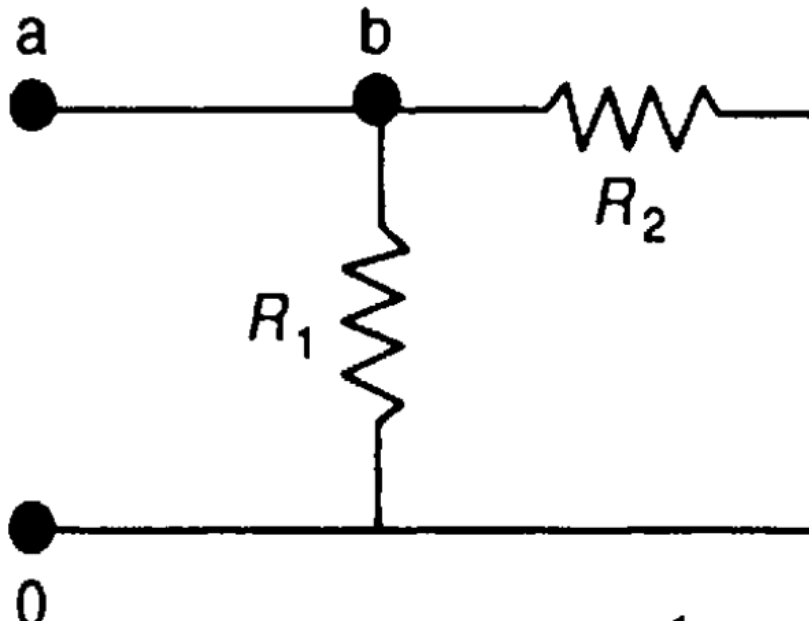


$$R = R_1 + R_2$$

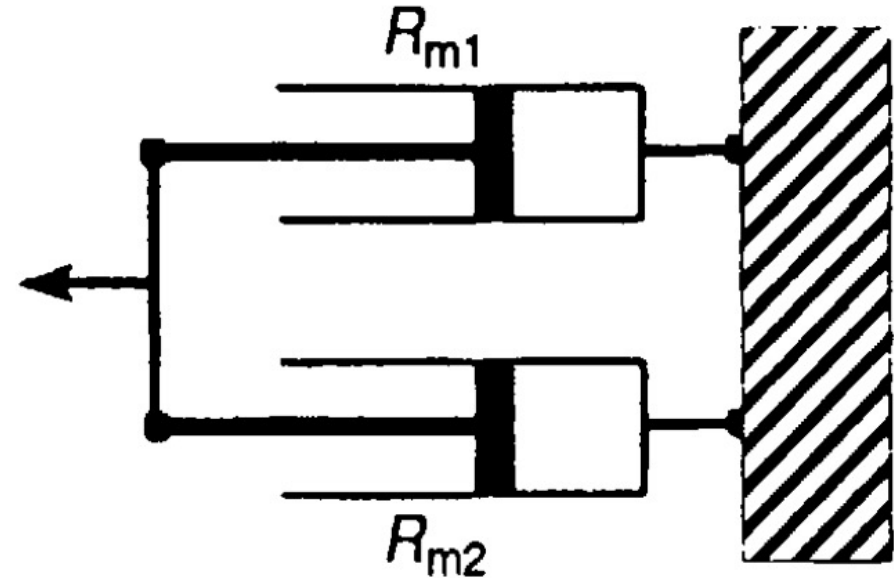


$$R = \left(\frac{1}{R_{m1}} + \frac{1}{R_{m2}} \right)^{-1}$$

PARALLEL COMBINATIONS OF ELECTRICAL RESISTANCE AND MECHANICAL DASHPOT

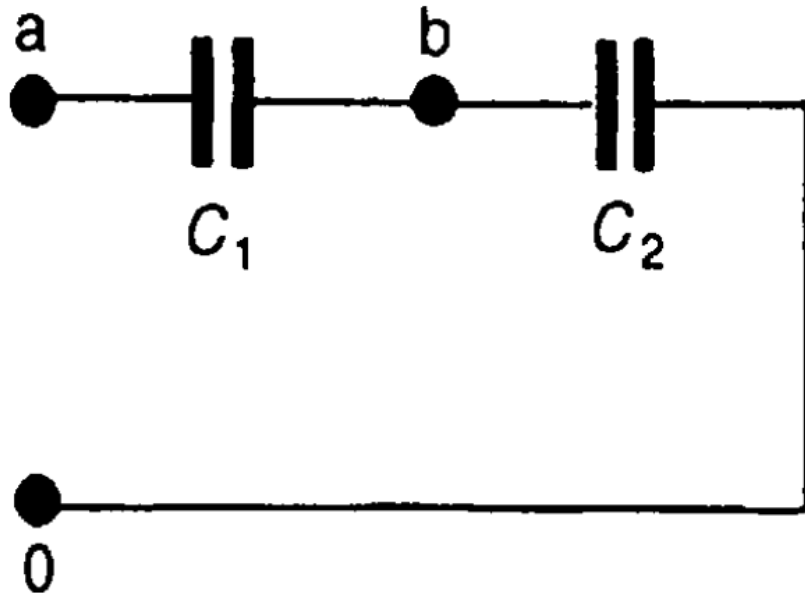


$$R = \left(\frac{1}{R_1} + \frac{1}{R_2} \right)^{-1}$$

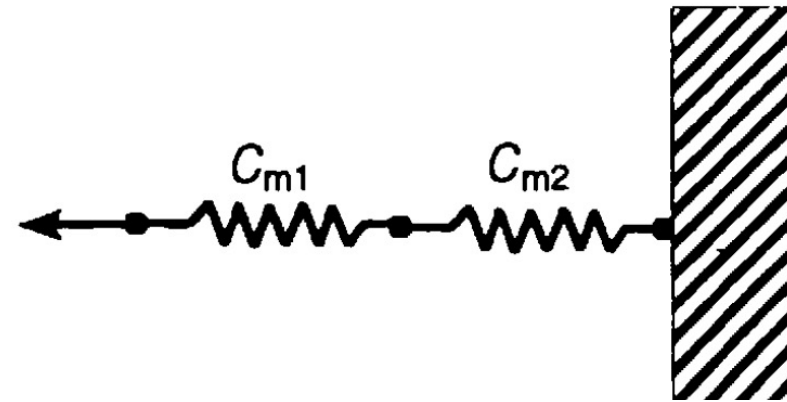


$$R = R_{m1} + R_{m2}$$

SERIES COMBINATIONS OF ELECTRICAL CAPACITANCE AND MECHANICAL SPRINGS

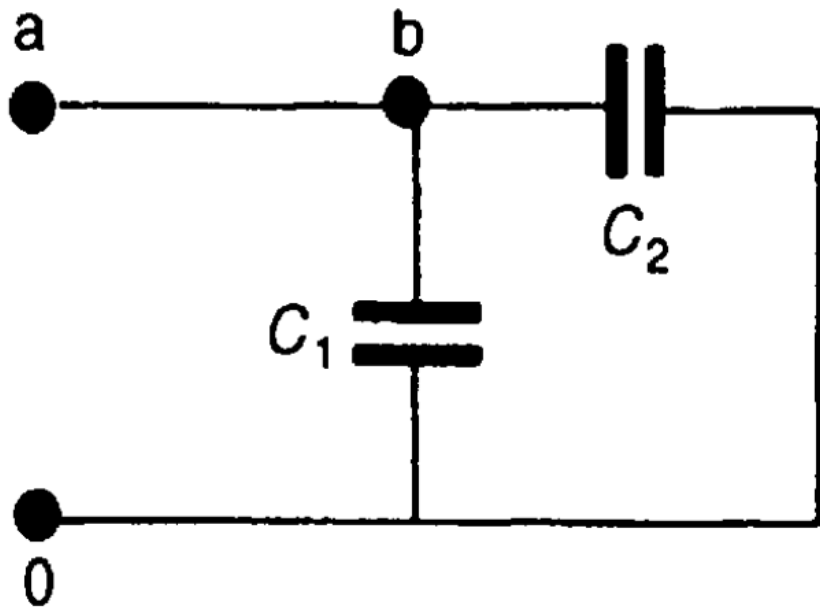


$$C = \left(\frac{1}{C_1} + \frac{1}{C_2} \right)^{-1}$$

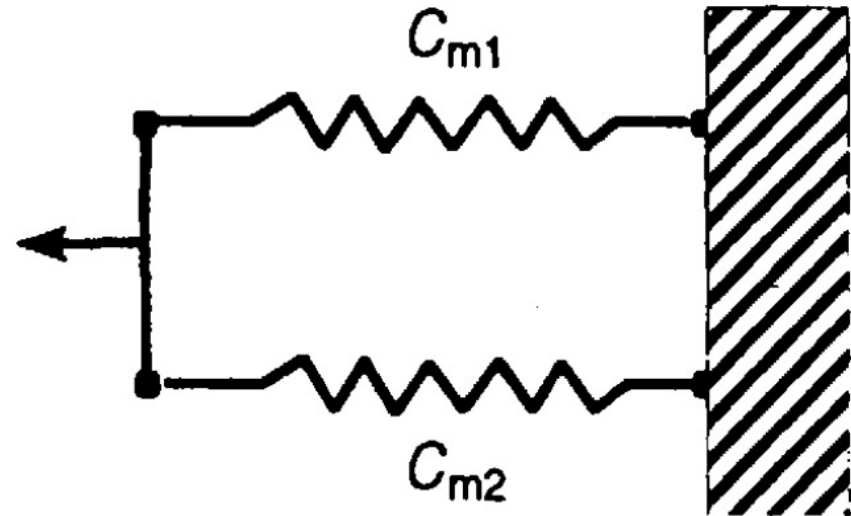


$$C = C_{m1} + C_{m2}$$

PARALLEL COMBINATIONS OF ELECTRICAL CAPACITANCE AND MECHANICAL SPRINGS

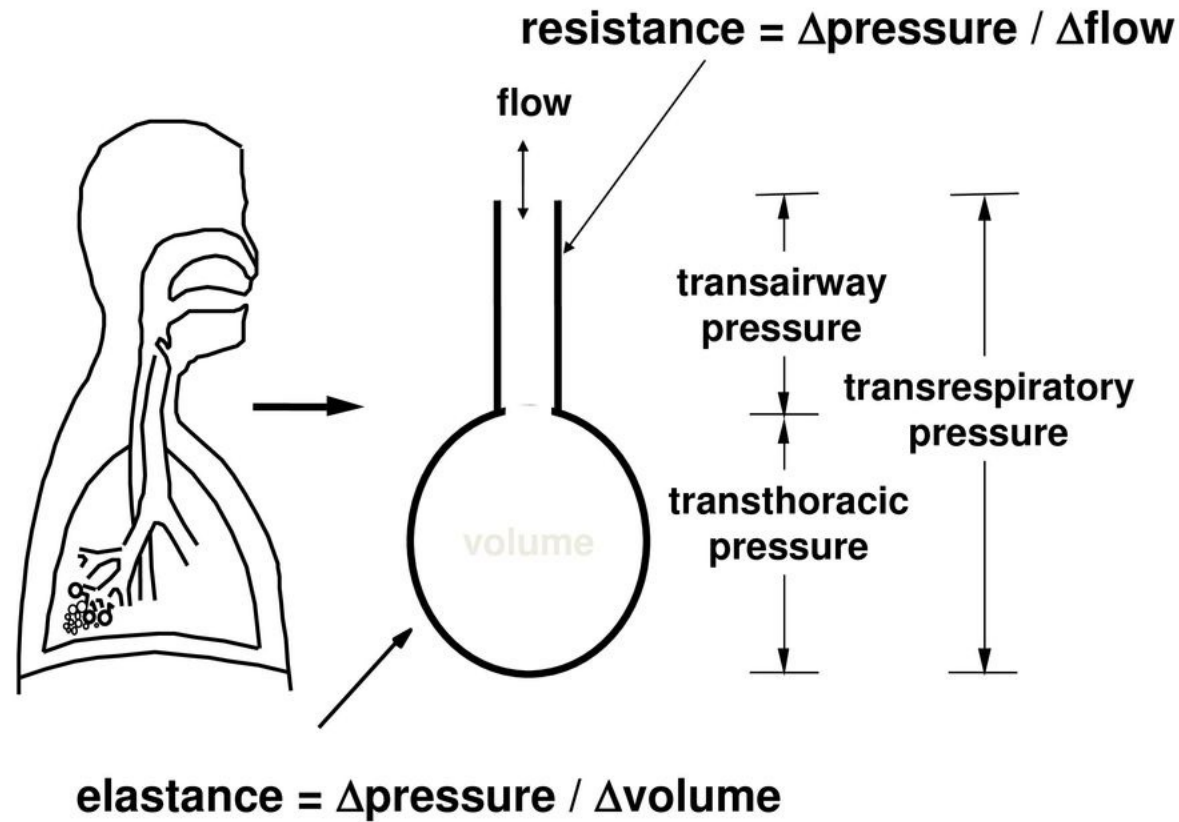


$$C = C_1 + C_2$$



$$C = \left(\frac{1}{C_{m1}} + \frac{1}{C_{m2}} \right)^{-1}$$

LINEARIZED RESPIRATORY MECHANICS

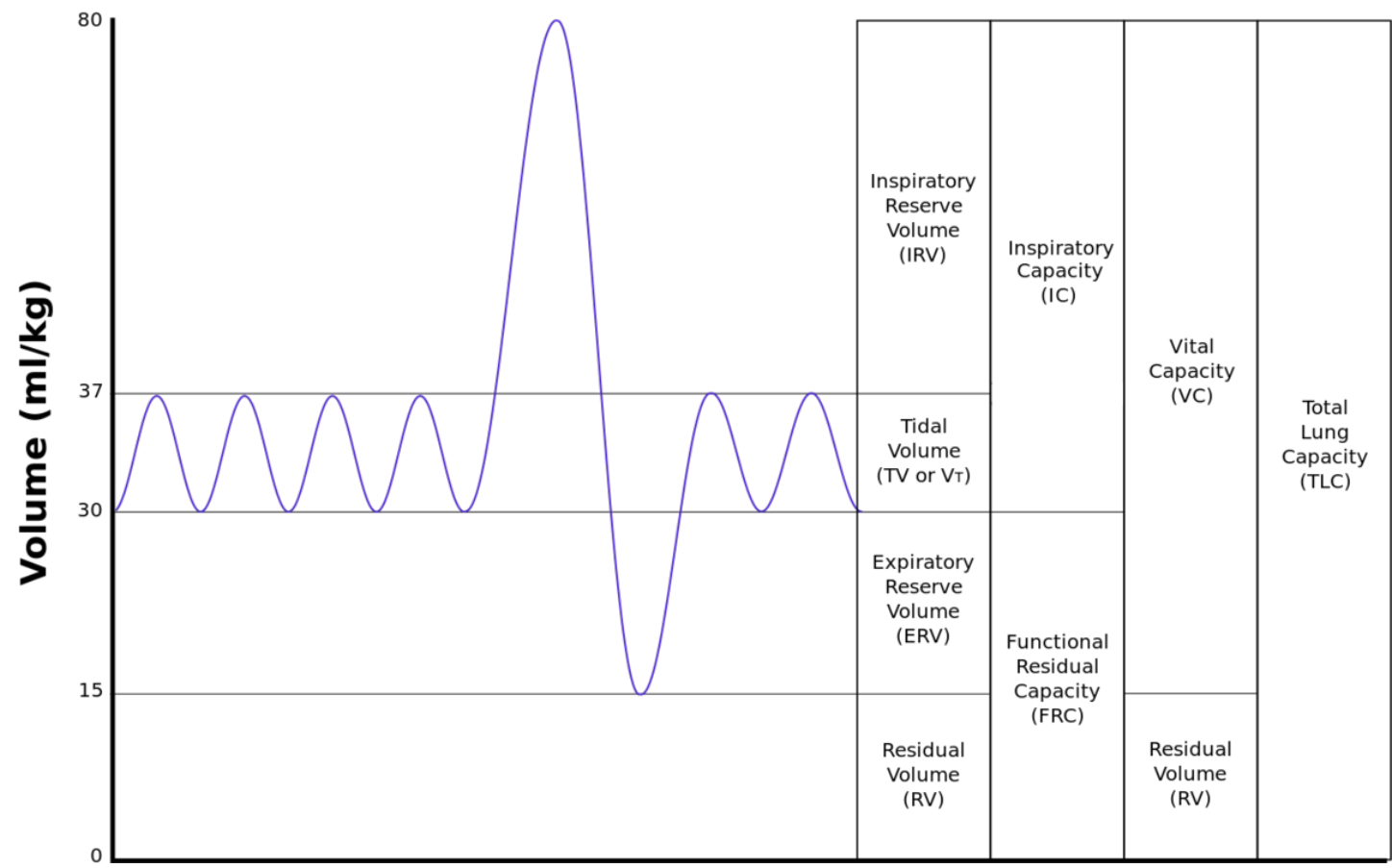


- **Respiratory Mechanics:** Encompasses the entire respiratory system, including the lungs, chest wall, and muscles involved in breathing. It provides a comprehensive view of how these components work together to generate airflow and maintain ventilation.
- **Lung Mechanics:** Focuses specifically on the lungs, their compliance, airway resistance, and elastic properties, without incorporating the chest wall or respiratory muscles. It is used to understand lung-specific issues affecting breathing.

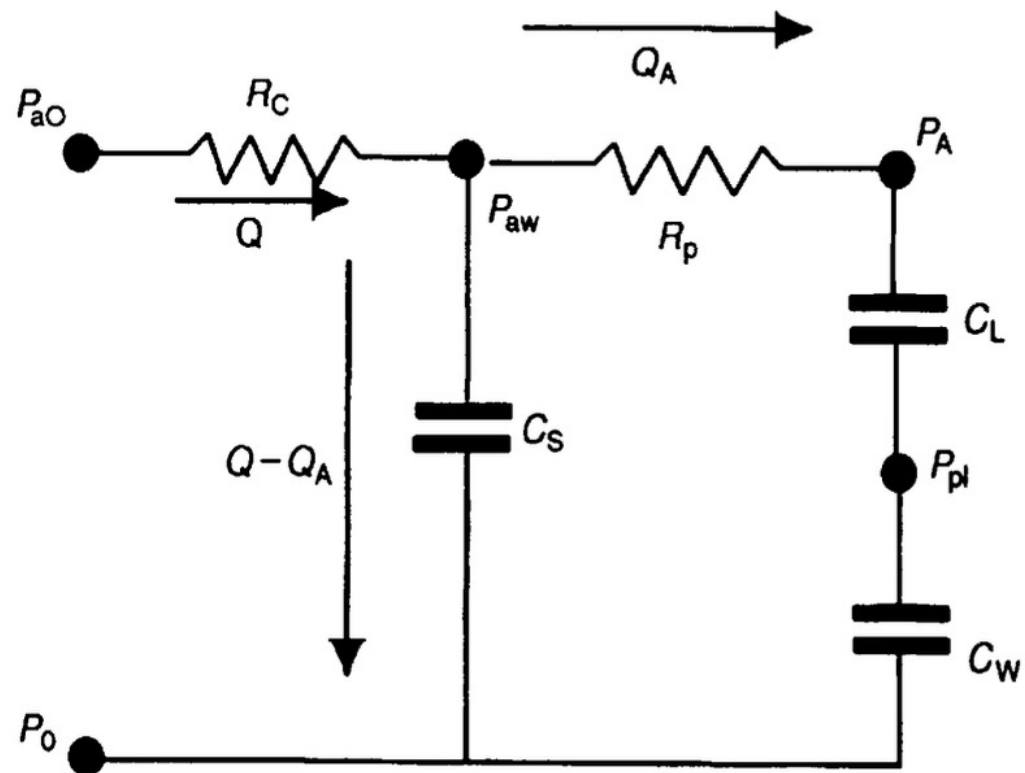
- Airflow: air moves from high-pressure to low-pressure areas within the respiratory system, driven by changes in thoracic volume.
- Transpulmonary Pressure: pressure difference between the alveolar pressure (inside the lungs) and the pleural pressure (in the chest cavity).
- Airway Resistance: opposition to airflow caused by the airways. It is influenced by factors such as airway diameter, length, and the viscosity of air. High resistance implies that greater pressure is needed to achieve a certain airflow.

- Lung Compliance: refers to the ease with which the lungs expand in response to pressure changes. High compliance indicates that the lungs are more easily stretched (e.g., in emphysema), while low compliance means the lungs are stiffer (e.g., in pulmonary fibrosis).
- Elastance: inverse of compliance and reflects the lungs' tendency to return to their original shape after being stretched. A high elastance means the lungs are more resistant to stretching, which often correlates with low compliance.

- Tidal Volume: volume of air inhaled or exhaled with each breath during normal breathing.
- Residual Volume: volume of air remaining in the lungs after a full exhalation.
- Functional Residual Capacity: volume of air left in the lungs at the end of a normal, passive exhalation.
- Total Lung Capacity: maximum volume of air the lungs can hold, including residual volume.



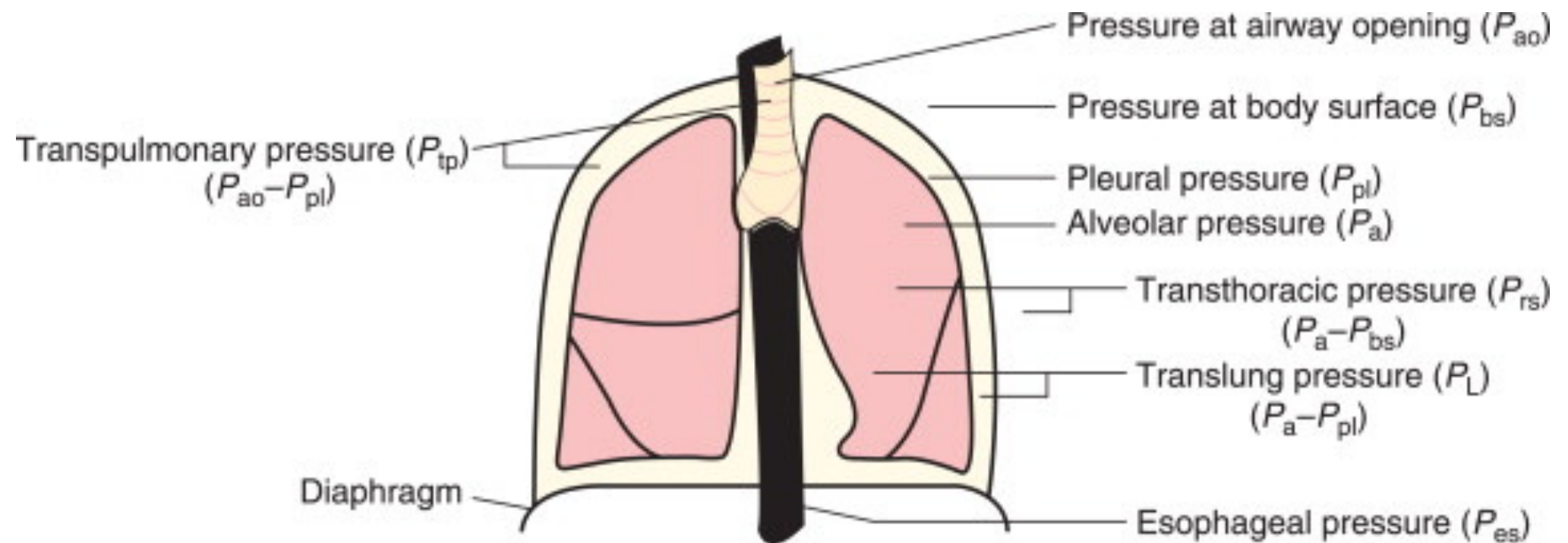
LINEAR MODEL



- The airways are divided into two categories: the larger or central airways and the smaller or peripheral airways, with fluid mechanical resistances equal to R_c and R_p , respectively.
- Air that enters the alveoli also produces an expansion of the chest-wall cavity by the same volume. This is represented by the connection of the lung (C_L) and chest-wall (C_w) compliances in series.

- However, a small fraction of the volume of air that enters the respiratory system is shunted away from the alveoli as a result of the compliance of the central airways and gas compressibility.
- This shunted volume is very small under normal circumstances at regular breathing frequencies, but becomes progressively more substantial if disease leads to peripheral airway obstruction (i.e., increased R_p) or a stiffening of the lungs or chest-wall (i.e., decreased C_L or C_W).
- this effect is accounted by placing a shunt compliance, C_S , in parallel with C_L and C_W .

- The pressures developed at the different points of this lung model are:
 - P_{ao} at the airway opening,
 - P_{aw} in the central airways,
 - P_A in the alveoli and
 - P_{pl} in the pleural space (between the lung parenchyma and chest wall).
- These pressures are referenced to the ambient pressure P_o (set to zero).



- Let the volume flow-rate of air entering the respiratory system is Q .
- In the linear model of respiratory mechanics, the primary relationship between pressure (P), airflow (Q) and resistance (R) is described by Ohm's law:

$$\Delta P = Q * R$$

- Transfer Function: derive a mathematical relationship between P_{ao} and Q .
- From Kirchhoff's Second Law (applied to the node P_{aw}), if the flow delivered to the alveoli is Q_A then the flow shunted away from the alveoli must be $(Q - Q_A)$.

- Applying Kirchhoff's First Law to the circuit containing R_C and C_S :

$$P_{a0} = R_C Q + \frac{1}{C_S} \int (Q - Q_A) dt$$

- Applying, Kirchhoff's First Law to the closed circuit containing C_S , R_P , C_L , and C_W :

$$R_P Q_A + \left(\frac{1}{C_L} + \frac{1}{C_W} \right) \int Q_A dt = \frac{1}{C_S} \int (Q - Q_A) dt$$

- Differentiating the Equations with respect to time, and subsequently reducing the two equations to one by eliminating Q_A , the equation relating P_{ao} to Q :

$$\frac{d^2 P_{ao}}{dt^2} + \frac{1}{R_P C_T} \frac{dP_{ao}}{dt} = R_C \frac{d^2 Q}{dt^2} + \left(\frac{1}{C_S} + \frac{R_C}{R_P C_T} \right) \frac{dQ}{dt} + \frac{1}{R_P C_S} \left(\frac{1}{C_L} + \frac{1}{C_W} \right) Q$$

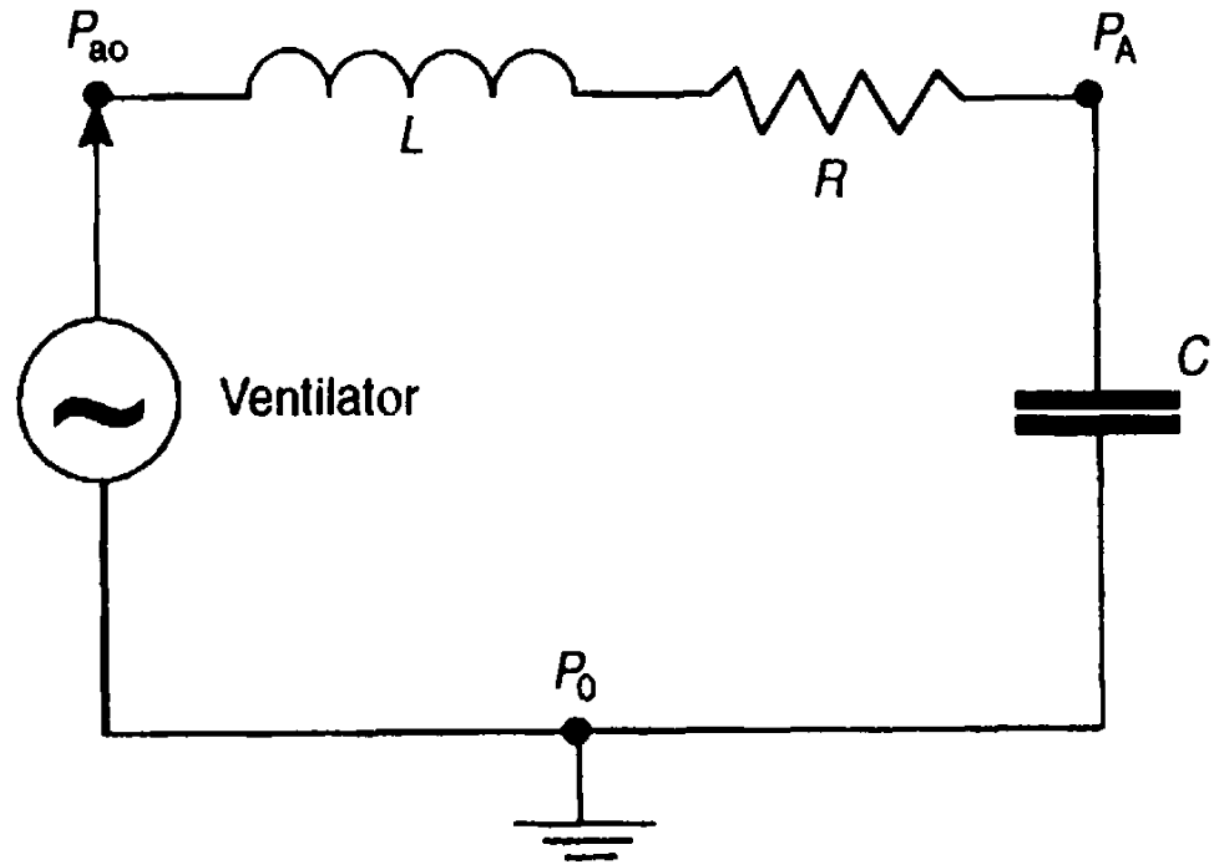
$$C_T = \left(\frac{1}{C_L} + \frac{1}{C_W} + \frac{1}{C_S} \right)^{-1}$$

- Taking Laplace Transform:

$$\left(s^2 + \frac{s}{R_P C_T}\right) P_{a0}(s) = \left(s^2 R_C + s \left(\frac{1}{C_S} + \frac{R_C}{R_P C_T}\right) + \frac{1}{R_P C_S} \left(\frac{1}{C_L} + \frac{1}{C_W}\right)\right) Q(s)$$

$$\frac{Q(s)}{P_{a0}(s)} = \frac{\left(s^2 + \frac{s}{R_P C_T}\right)}{\left(s^2 R_C + s \left(\frac{1}{C_S} + \frac{R_C}{R_P C_T}\right) + \frac{1}{R_P C_S} \left(\frac{1}{C_L} + \frac{1}{C_W}\right)\right)}$$

SIMPLIFIED VERSION



- R represents a combination of resistance to airflow in the airways, lung tissue resistance, and chest-wall resistance.
- C represents the combined compliance of lung tissue, chest wall, and airways.
- L represents fluid inertance in the airways.
- Q represents the airflow rate.
- Let predict how the alveolar pressure (P_A) will respond dynamically to different pressure waveforms (P_{a0}) applied at the airway opening.

- Applying Kirchhoff's First Law to the model, the pressure drop across the entire model must be equal to the sum of all the pressure drops across each of the circuit elements.

$$P_{a0} - P_0 = L \frac{dQ}{dt} + RQ + \frac{1}{C} \int Q dt$$

- similar expression can be derived to relate P_A to Q :

$$P_A - P_0 = \frac{1}{C} \int Q dt$$

- reference all pressures to the ambient pressure (i.e., set $P_0 = 0$).
- combining and eliminating Q from both equations,

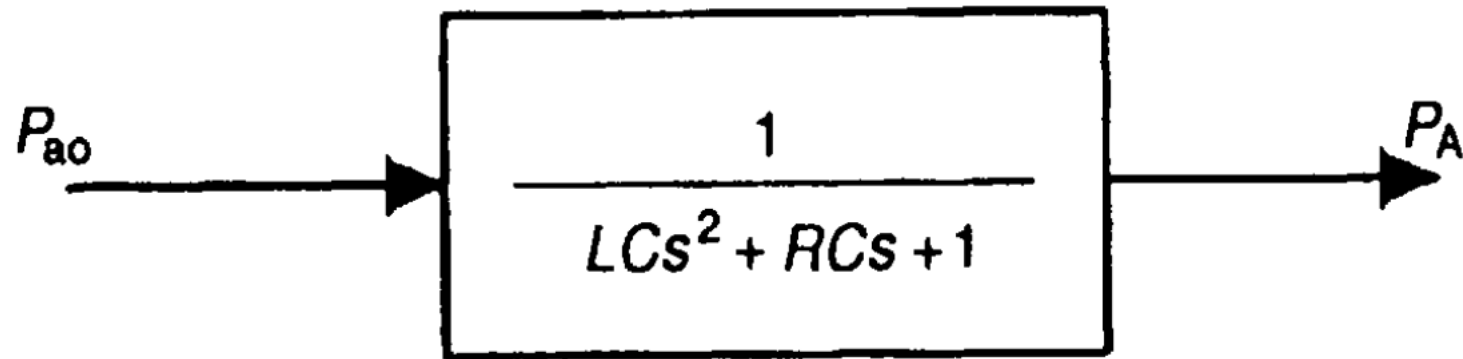
$$P_{a0} = LC \frac{d^2 P_A}{dt^2} + RC \frac{dP_A}{dt} + P_A$$

- Taking Laplace Transform:

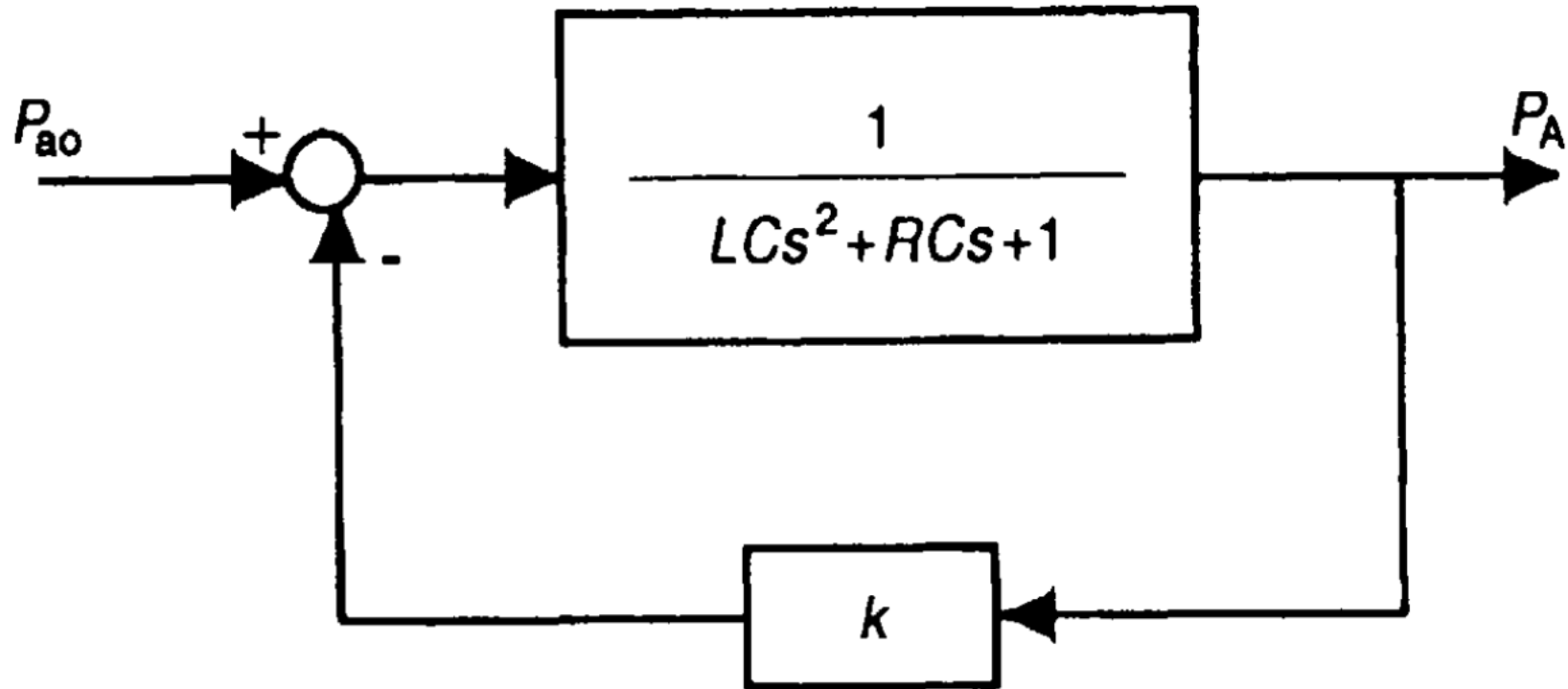
$$P_{a0}(s) = (LCs^2 + RCs + 1)P_A(s)$$

$$\frac{P_A(s)}{P_{a0}(s)} = \frac{1}{(LCs^2 + RCs + 1)}$$

OPEN-LOOP CONFIGURATION



CLOSED-LOOP CONFIGURATION



- In clinical setting, since large fluctuations in P_A can cause pulmonary barotrauma or damage to lung tissue.
- In order to control P_A , it is necessary to measure this variable and feed the measurement back to the controller.
- In practice, this can be achieved by measuring the pressure in the mid-esophagus with the use of an esophageal balloon, since fluctuations in esophageal pressure have been demonstrated to closely reflect fluctuations in alveolar pressure.

- The type of control scheme is known as proportional feedback, since the feedback variable is proportional to the system output.

$$\frac{P_A(s)}{P_{a0}(s) - kP_A(s)} = \frac{1}{(LCs^2 + RCs + 1)}$$

- Overall transfer function of the closed-loop system:

$$\frac{P_A(s)}{P_{a0}(s)} = \frac{1}{(LCs^2 + RCs + (1 + k))}$$

- Generalized transfer function to represent both the open-loop and closed-loop conditions:

$$\frac{P_A(s)}{P_{a0}(s)} = \frac{1}{(LCs^2 + RCs + \lambda)}$$

- where $\lambda = 1$ for the open-loop case, and $\lambda = (1+k)$ for the closed-loop case.